The Value of Stopping: Evidence Queues, Recoverability, and Priority Rights

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Abstract

Correlated evidence can make efficient research privately unattractive both initially and at continuation. In a two-case, two-slot model with expiration and optional terminal action, we derive the exact implementation and funding frontier for two nonnegative public-history credits. One rewards the initial experiment; the other rewards a specified contingent final experiment. Complete backward induction gives necessary and sufficient inequalities, separate expected and pathwise gross-funding limits, and attainment boundaries on a certified interval of binary priors. Credits are transfers rather than resource losses. The independent benchmark retains the recoverability-versus-priority comparison and common-charge classification. Complete proofs and exact synthetic enumeration support this restricted implementation result; no unrestricted mechanism optimum or empirical effect is claimed.

Keywords: evidence acquisition; scarce attention; decision queues; stopping; priority rights; expiration.

1. Introduction

A request for one more analysis can be justified for its own decision and still be costly for the organization. The analyst may have to postpone research on another case whose purchasing quote expires sooner, whose decision remains unusually uncertain, or whose recoverable value will be lost if action is delayed. The relevant opportunity cost is therefore endogenous to the competing case. It is not generally a fixed price for elapsed time.

We study two evidence cases sharing one analyst for two research slots. Each case can be acted on immediately without new evidence, or receive one pre-specified experiment before action. While a case is kept unresolved, holding costs accrue and its opportunity may expire. A case assigned the second slot can instead be closed immediately. Thus the allocation of attention and the decision to stop researching must be determined jointly. The experiment is a bounded evidence packet, not an open-ended search process. This finite restriction makes the organizational equilibrium, corrective rule, and implementation boundaries fully solvable.

For case i, let b_i be the net gain from immediate research rather than immediate action. Let w_i be the optimal gain from being offered research only after one waiting slot, allowing immediate action instead. Then $u_i = b_i^+ - w_i$ is the loss from delaying that research option. The efficient order compares u_i, not just b_i. This loss depends on beliefs, signal precision, stakes, recoverability, holding costs, and expiration.

We analyze a concrete decentralized institution: case A initially owns the right to the first slot. Its owner can claim that slot, release it and wait, or act without research. Case B then receives the available slot; final-slot choices are privately optimal. Owners value their own case's payoff and pay their own research and holding costs. Even with complete information and no split accounting of these direct costs, A generally claims early service whenever its own research has positive net value. It ignores how much B loses by waiting.

The independent benchmark contributes a recoverability-versus-deferral-priority comparison and the exact restrictions under which a common research charge can implement efficient retention and order. The underlying scheduling interchange and externality toll are classical. We give a proposition-by-proposition comparison rather than claiming a new Gittins index or

dynamic mechanism. We cannot certify that the bounded increments are unoccupied because one close predecessor remains unavailable in full text.

We therefore also solve the smallest adjacent nonseparable margin, still with two cases, one experiment each, and two slots. An explicitly correlated prior makes the first public signal change the other case's research value. A previously researched case may remain unresolved to benefit from the second signal. This creates a cross-information benefit as well as congestion. We derive all contingent joint choices, the corresponding finite local game, and the exact final-slot boundary at which an information credit rather than a positive toll is needed. The central correlated result then closes the initial incentive gap using just two observable-history credits. We solve the complete local game, give exact necessary and sufficient implementation inequalities, and distinguish expected funding from pathwise capacity, including equality and nonattainment. An exact interval of priors around the original witness makes this a parameter-region result. Neither correlation nor conditional subsidies are claimed as new principles.

2. Literature and contribution boundary

Henry and Ottaviani (2019) study costly sequential research under divided control and conflicting terminal interests. Escobar and Zhang (2021) analyze private learning, communication, and prompt implementation in dynamic delegation. They already establish that organizational rules can alter learning and delay. We instead align owners with their own case's economic payoff, make primitives public, and introduce competition between cases for a single analyst. Our friction does not require persuasion, private signals withheld from the organization, or different action preferences within a case.

Sequential sampling is the separable foundation, not a novelty claim. Fudenberg, Strack, and Strzalecki (2018) connect costly information gathering to decision timing. Xiao, Hu, and Wang (2026), published online in *Management Science*, characterize recursive standards in delegated finite search. Their publisher abstract and publication status were verified; full technical access remained unavailable. We cannot certify priority against every result in that paper.

The relevant scheduling comparison is broader than sequential sampling. Gittins (1979) develops dynamic allocation indices, with applications to scheduling and search; Weber (1992) provides a proof of the classical index policy. Those results are much more general than our two-job calculation in important respects. Our u_i is a finite pairwise interchange score, not a renamed Gittins benefit/time ratio. Inactive evidence cases can expire and the common finite horizon shrinks, so a general rested-bandit theorem cannot simply be invoked to justify an index policy for extensions of our model.

Expiring-arm and dynamic-pivot precedents are mapped explicitly in Sections 2.1-2.3. Whittle (1988), Weber and Weiss (1990), Righter and Shanthikumar (1998), and Yu, Xu, and Tong (2017 preprint; later 2018 journal publication) are scheduling comparators. Adiri and Yechiali (1974) and Bergemann and Välimäki (2010) are pricing precedents. The posted public-state toll is not claimed to duplicate a complete private-information pivot mechanism.

Campbell, Ederer, and Spinnewijn (2014) study effort and communication around deadlines. Their mechanism cautions against interpreting our no-effort-response deadline result as universal. Eyring, Ferguson, and Koppers (2021) vary the composition of performance feedback, not acquisition or queue priority. Their experiment supports separating display from acquisition interventions. Xu, Zariphopoulou, and Zhang (2026, arXiv v4) study reversible choices with changing information regimes. Our recoverability parameter is more limited: it modifies the wrong-action payoff without creating post-action learning.

The primary source ledger records the depth of reading and version differences. Full text was read for several close organizational and pricing papers and the deadline-scheduling preprint; original Gittins, Righter-Shanthikumar, and recent delegated-search technical coverage remains

incomplete. We make no exhaustive-literature or established-priority claim.

2.1 Primitive mapping to expiring-arm scheduling

Our independent-case state is (p_i, e_i, l_i, t): belief, remaining experiment flag, live/retired status, and the common clock. An active arm pays $-c_i - d_i$, faces expiry, and if surviving generates its one signal. A passive retained arm pays $-d_i$, faces expiry, and keeps its belief. Retirement pays $g_i(p_i)$ once and removes the case; expiry pays zero. Thus this is a finite-horizon expiring-arm stopping problem. Neither expiry nor the optional stopping action is a new scheduling principle.

Subtracting the potential g_i makes the correspondence precise. The one-slot expected transformed reward is b_i when active, $-\ell_i$ when passive, and zero when retired. The initial potential is added back once. This is just accounting; it does not identify a new index or prove indexability. The clock and passive expiry remain state changes even when the belief is frozen. The optional immediate payoff is not a reason to declare the problem outside every existing bandit framework: it can be encoded as a terminal action. Time augmentation likewise accommodates a finite horizon, although it need not preserve a rested-arm index theorem's assumptions.

Righter-Shanthikumar's publisher abstract confirms index results under independent expiry. A lawful full version was not retrieved after checking the publisher and both authors' public bibliographies. We did **not** inspect their exact active/passive hazard assumptions, arm transition/reward restrictions, horizon/discount criterion, retirement options, information structure, or index optimality conditions on pages 453-468. None of those cells is marked verified below. We therefore make no claim that their theorem excludes or includes our model. An abstract cannot supply that comparison.

Whittle's publisher description concerns relaxed capacity prices and active/passive evolution. The author-hosted Weber-Weiss full text establishes that asymptotic index conclusions require additional conditions. The read Yu-Xu-Tong v5 preprint treats stochastic deadlines, capacity, costs and noncompletion penalties; it is not the same read object as its later journal publication. These precedents establish occupied problem classes, not a theorem-by-theorem novelty gap for this paper.

2.2 Primitive mapping to dynamic-pivot pricing

Bergemann-Välimäki's published Section 3 allocates a central facility, one task per period, with delay represented by discounting. Section 4, equations (5)-(7), prices current and continuation externalities using programs with and without an agent. Their independent private-value Markov framework yields truth-telling and participation results that are not proved here.

For our two independent routes, B's optimal incremental value if A releases first service is b_B^+; if A occupies it, B retains or acts and obtains w_B. Their difference u_B is exactly the marginal externality algebra, with the common baseline g_B canceling. There is no extra loss to add after subtracting those two values. Our asymmetric right-holder charge and uncharged B are not the complete dynamic-pivot transfer vector. Internal charges are excluded from firm resource payoff. A finite absorbing horizon is not, by itself, an economic novelty relative to dynamic pricing; finite flows can be represented separately from a discount convention.

In the correlated extension, one case's signal changes the other's posterior. The independently evolving private-value conditions in that published theorem are not directly satisfied. This is a boundary of that direct theorem mapping, not a claim that correlated-information externality pricing is absent from the wider literature. The extension keeps information public and makes no private-type truthfulness claim.

The restricted credits in Sections 8.4-8.5 do not replicate realized team payoffs or the full dynamic-pivot payment vector. The published pivot equations compute each agent's flow and

continuation externality from with/without-agent programs; our two instruments are constrained to specified history/action events. Their independent private-value truthfulness theorem is not being extended to correlated private types. Known public beliefs, observable research and enforceable credit commitment suffice here; eventual action-payoff observability is unnecessary. Their positive payment variable is subtracted from the agent's utility; our credits enter with the opposite sign. Their published no-deficit conclusion therefore does not supply our gross credit budget. The funding frontiers describe internal appropriation authority, not budget balance or the pivot mechanism's revenue properties.

Moroni (2022), published introduction and model, studies history-dependent incentives and free-riding in a multistage risky project with privately observed effort and learning, limited liability and endogenous assignments. That precedent further rules out novelty claims for conditional experimentation rewards. Our observable one-experiment actions, two scarce slots and two fixed payment events define a different restricted calculation; we do not infer an unrestricted contracting advantage from that difference.

2.3 Result-by-result comparison

Result in this paper	Closest established principle	Exact bounded claim and remaining limit
Separable benchmark, Section 3	Standard sequential stopping and benefit/time accounting	No novelty claimed
Theorem 1	Finite two-job interchange; expiring-arm scheduling	Explicit optional retirement/retention solution; not a new index; Righter-Shanthikumar conditions uninspected
Theorem 2 and exact fee set	Local queue priority and restricted pricing	Necessary/sufficient common-fee intervals within the stated institution and tie convention; no claim all price mechanisms fail
Corollary 1	Enumeration of feasible deadlines	Exact three-deadline comparison only
Theorem 3 and Corollary 2	Marginal externality and priority pricing	Public-state asymmetric implementation; not a new dynamic-pivot theorem
Theorem 4	Bayes-risk payoff comparison combined with deteriorating-option scheduling	Opposite recoverability derivatives on the retained-research region; priority crossing is conditional and priority remains uncertified
Theorems 5-6	Finite Bayesian dynamic programming; information externalities	Correlated joint solution and final-slot boundary, retained benchmarks

Result in this paper	Closest established principle	Exact bounded claim and remaining limit
Theorems 7-8	Conditional subsidies, free-riding compensation, restricted dynamic transfers	Complete two-history-credit implementation and piecewise expected/pathwise funding frontier on I; no unrestricted optimal-contract or pivot claim

Klein and Rady (2011) already study strategic experimentation with correlated risky types. The June 23, 2010 author-hosted manuscript and 2011 publisher metadata were checked; their continuous-time two-player experimentation setting is distinct from our two-slot retirement problem. Correlation and cross-learning themselves are not claimed new. The inspected working model permits simultaneous safe/risky choices with publicly observed payoffs and continuous-time learning, rather than our single analyst, two optional one-shot experiments and expiring action opportunities. Its public cross-learning and free-riding logic is inherited. Theorems 7-8 instead restrict the payment support to two existing observable research histories, preserve local stopping rights, and solve their funding limits. No located theorem was verified to state this exact restricted frontier; that bounded reading is not an exhaustive priority certification. The uninspected independent-expiry article is not a positive premise of the correlated-credit result.

3. Separable stopping benchmark

To locate the new mechanism, consider first a single finite evidence tree. At live history x, immediate action pays $g(x) = \max_a E[u(a,\theta) \mid \mathcal{F}_x]$. An experiment costs c_e, consumes a period with additive delay price d, and survives with probability s_e. The filtration records observed signals, research choices, elapsed slots, and opportunity survival. Feasible policies are adapted and terminate at a finite horizon. Expiration pays zero. The value obeys

$$V_d(x) = \max\{g(x), \max_e[-c_e - d + s_e E_e V_d(x')]\}.$$

For any adaptive policy π that continues first, define expected terminal reward R_π, acquisition cost C_π, and active periods $T_\pi > 0$. Then continuing is strictly better than acting now if and only if

$$d < D(x), \qquad D(x) = \max_\pi \frac{R_\pi - C_\pi - g(x)}{T_\pi}.$$

This follows by rewriting each policy's incremental payoff as $R_\pi - C_\pi - g - dT_\pi$. It is elementary finite optimal stopping. If an analyst internalizes only αd, its continuation region uses $\alpha d < D(x)$. Charging the missing cost aligns objectives only when this marginal charge enters actual preferences and work cannot escape the ledger.

Likewise, Bellman action regret telescopes: optimal value minus any feasible policy's payoff equals the expected sum of optimal action-value gaps along that policy's path, including expiration. The complete proofs and earlier exact single-case illustrations are retained in Appendix B. They are not the revised paper's central contribution.

A bound $d \in [d_L, d_U]$ provides a stopping certificate only conditional on the full payoff, signal, expiration, action, and policy model. It is not robustness to misspecified likelihoods or missed alternatives. In the shared-resource models below, no exogenous d alone represents the forgone value of another case.

4. Two cases sharing one analyst

4.1 Timing, primitives, and economic accounting

There are two cases $i \in \{A, B\}$, independent latent binary states θ_i, and public priors p_i. Slots are $t = 0, 1$, followed by mandatory terminal action on surviving unresolved cases at $t = 2$. One analyst can perform at most one experiment in each slot. Each case has at most one experiment. Its binary signal has accuracy $q_i \in [1/2, 1]$, conditionally independent of the other case. All parameters and available actions are public.

At each slot boundary the organization may act on any subset of unresolved cases, with no analyst time required for that terminal action. Correct action yields $S_i > 0$; wrong action yields $r_i S_i$, $0 \le r_i \le 1$. A resolved case leaves permanently. A case retained unresolved over a slot incurs $d_i \ge 0$ and survives the slot with independent probability $s_i = 1 - h_i$, where $0 \le h_i < 1$. If the analyst researches case i, it also incurs acquisition cost $c_i \ge 0$. The signal is received only if the case survives; the researched surviving case can act immediately at the next boundary. Since it has no experiments left, additional waiting is weakly dominated by action.

Expiration pays zero. Costs for an occupied slot are paid before expiration. Thus the same accounting applies whether a case is active or passively waiting. The additive holding cost is a residual resource or operating loss, excluding any loss already counted through the terminal payoff and expiration. A first-slot research on A with B retained costs $c_A + d_A + d_B$, not c_A twice and not a second artificial queue-loss penalty. Internal corrective transfers are not firm resource costs.

The filtration contains the public priors, decisions to stop or wait, analyst assignment, and observed survival and signals. An unrestricted joint policy may condition its final-slot assignment on all this information. Idle policies are feasible and included in the independent enumeration. The theorem will show that no idle policy improves on the stated solution. There are no new arrivals, strategic suppliers, correlated information, hidden effort, or endogenous experimental precision.

4.2 Information value and the option to retire

Write $m_i = \min(p_i, 1 - p_i)$ and let $\bar{m}_i$ be expected posterior Bayes error after the one available signal. Define

$$g_i = S_i[1 - (1 - r_i)m_i], \quad M_i = S_i[1 - (1 - r_i)\bar{m}_i],$$

$$b_i = s_i M_i - c_i - d_i - g_i, \qquad \ell_i = d_i + h_i g_i. \tag{1}$$

Here g_i is immediate-action value, M_i the expected payoff after observing the signal conditional on survival, and b_i the net gain from researching in the next available slot. The quantity ℓ_i is the expected loss from keeping the current decision unresolved for one slot even without acquiring evidence.

If case i is offered only the second slot, it may act now or wait and, conditional on survival, choose whether to research. Its incremental value relative to immediate action is

$$w_i = [s_i b_i - \ell_i]^+, \qquad z^+ = \max(z, 0). \tag{2}$$

Since $s_i \le 1$ and $\ell_i \ge 0$, $0 \le w_i \le b_i^+$. Define the option's deferral loss

$$u_i = b_i^+ - w_i \ge 0. \tag{3}$$

This loss includes optimal stopping. In particular, when deferred research is no longer worthwhile, $w_i = 0$ and $u_i = b_i^+$: displacing that case removes its entire positive research option, but it can still act now. It is incorrect to add its immediate payoff g_i as a second loss after using this expression.

5. Optimal joint stopping and scheduling

Theorem 1 (independent two-case solution). Let $G = g_A + g_B$. The optimal expected total payoff is

$$W^* = G + \max\{0,\ b_A + w_B,\ b_B + w_A\}. \tag{4}$$

If both $b_i > 0$, researching A first is optimal rather than researching B first exactly when $u_A \geq u_B$. After assigning i first, the other case j waits precisely when $s_j b_j > \ell_j$; otherwise it acts immediately. A surviving second-slot case is researched if $b_j > 0$, otherwise acted on. After the single signal, the researched case selects its posterior-optimal terminal action.

Proof. If case i is researched first, its expected incremental payoff is b_i. Independence implies that the other case learns nothing about its own state from i's signal or expiration. Its best alternative to immediate action is one slot of waiting followed by its own optimal research-or-act choice. That incremental payoff is w_j by (2). Therefore the best value conditional on i first is $G + b_i + w_j$. Stopping both gives G.

An idle first slot leaves at most one later experiment. Each retained case loses ℓ_i from baseline while waiting. If at most one retained case survives, its later research gain cannot exceed its immediate positive research gain. If both survive, the final allocation selects at most one. Removing the idle slot and performing that chosen experiment immediately is no worse in expectation: the expected maximum of the gains of surviving cases is at most $\max(b_A^+, b_B^+)$, and waiting losses are nonnegative. Immediate research on a maximizer and immediate action on the other case attains $G + \max(b_A^+, b_B^+)$. Hence idle policies cannot beat the candidates in (4). No other first-slot policy exists.

When both $b_i > 0$, subtracting the two research-order values gives $(b_A - w_A) - (b_B - w_B) = u_A - u_B$. The retention and final action rules follow from the same comparison. □

The formula is a finite scheduling interchange, not a general bandit index theorem. In this model signals affect posterior terminal action but do not create further research choices on the same case. Independence therefore eliminates informational cross-effects in the final schedule. The stochastic tree still includes both signal outcomes and both cases' expiration; a stronger multi-experiment model would generally need a joint Bellman solution and is not claimed solved here.

6. Decentralized priority rights and corrective policies

6.1 Local equilibrium

Case A owns the first-slot priority right. Its owner may research now, act now, or release the slot and wait while B chooses whether to research. If A claims the slot, B chooses whether to act now or wait for the second slot. Subsequent research and terminal actions are locally optimal. Owners maximize their own case's expected payoff; no owner receives the other's payoff. Internal transfers enter their objectives linearly. Ties between acting and costly research favor action; ties between using an available slot now and waiting with identical payoff favor earlier service. These conventions and public primitives define a perfect-information sequential game.

A common nonnegative charge τ is paid for every experiment, regardless of when it occurs, to a central budget. Let $\tilde{b}_i = b_i - \tau$ and

$$\tilde{w}_i = [s_i(b_i - \tau) - \ell_i]^+.$$

Theorem 2 (independent local equilibrium and common-charge boundary). Under the stated tie convention, if $b_A > \tau$, A researches first and B waits if and only if $\tilde{w}_B > 0$. If $b_A \le \tau$, A acts immediately and B researches if and only if $b_B > \tau$. When $\ell_A > 0$, A's preference for immediate research over waiting is strict whenever $b_A > \tau$.

Suppose the unique efficient order is B first, $b_A, b_B > 0$, and $\ell_A > 0$.

(a) If $w_A > 0$, no common $\tau \ge 0$ implements the efficient schedule and retention policy.

(b) If $w_A = 0$, every $\tau \in [b_A, b_B)$ implements it; this interval is nonempty.

If A first is efficient, $\tau = 0$ implements an efficient policy. If $b_A \le 0$, $\tau = 0$ lets B research when worthwhile. Thus failure is conditional, not universal.

Proof. With positive $\tilde{b}_A$, immediate research gives A gain $\tilde{b}_A$. If A releases the first slot, its best waiting gain when B uses it is $\tilde{w}_A \le \tilde{b}_A$. The difference before truncation is $h_A\tilde{b}_A + \ell_A$, positive if $\ell_A > 0$. If B does not use the released slot, idling provides no improvement. Hence A claims service. If $\tilde{b}_A \le 0$, waiting cannot help and A acts. B's behavior follows (2) with the charged research cost.

For (a), reversing A's strictly preferred first claim requires $\tau \ge b_A$. But then $b_A - \tau \le 0$, so A will not research in the last slot and cannot have positive retained research value. If $\tau < b_A$, A takes the first slot. Neither case gives the desired policy.

For (b), unique B-first optimality and $w_A = 0$ imply $b_B > b_A + w_B \ge b_A$. The displayed interval makes A act and B research. The other implementation claims follow from the equilibrium at $\tau = 0$. □

For an exact classification, let $t_B = b_B - \ell_B/s_B$. Excluding transfers from resource payoff, the equilibrium value at every $\tau \ge 0$ is

$$J_\tau = \begin{cases} G + b_A + \mathbf{1}\{\tau < t_B\} w_B, & \tau < b_A, \\ G + \mathbf{1}\{\tau < b_B\} b_B, & \tau \ge b_A. \end{cases} \tag{5a}$$

Thus the exact implementation set, including ties, is $\{\tau \ge 0 : J_\tau = W^*\}$. For a unique A-first optimum with $w_B > 0$, it is $[0, \min\{b_A, t_B\})$; with $w_B = 0$, it is $[0, b_A)$. For a unique B-first optimum with $b_A \le 0$, it is $[0, b_B)$; with $b_A > 0, w_A = 0$, it is $[b_A, b_B)$; with $b_A > 0, w_A > 0$, it is empty under the early-service tie convention. If both research gains are nonpositive, every nonnegative fee implements immediate action. The strict-urgency qualification below distinguishes the empty-set claim from an impossibility under all favorable alternative tie-breaking. Equation (5a) follows directly from the two equilibrium branches: a charge can change a choice, but the charge is not itself a social resource expense.

If $\ell_A = 0$, then $h_A = d_A = 0$ because $g_A > 0$. Early and late private service can tie. Under favorable alternative tie-breaking, an efficient B-first schedule with A retained can then be supported at zero charge. Our robust impossibility result requires strict local urgency, not an arbitrary unfavorable tie. A change in priority ownership or a dispatcher that already assigns the efficient order can likewise implement at zero charge. We are evaluating charges within the specified decentralized institution, not proving every constant-price mechanism impossible.

6.2 Common deadline

A common binding deadline leaves $D = 0, 1$, or 2 slots for both cases; it changes neither priors nor priority rights. At zero charge its equilibrium payoffs are

$$J_0 = G,$$

$$J_1 = \begin{cases} G + b_A, & b_A > 0, \\ G + b_B^+, & b_A \le 0, \end{cases}$$

$$J_2 = \begin{cases} G + b_A + w_B, & b_A > 0, \\ G + b_B^+, & b_A \le 0. \end{cases} \tag{5}$$

Corollary 1. If B first is uniquely efficient and $b_A > 0$, no unpriced common deadline implements the optimum. Conversely, if A first is efficient, $D = 2$ implements an efficient policy; if no research is valuable, $D = 0$ does so. Equation (5) gives the exact test including ties and other cases.

Proof. Under unique B-first optimality, $W^* > G + b_A + w_B \ge G + b_A > G$. These three quantities bound the payoffs of the three deadlines when A has positive research value. The converse statements follow from (4)-(5). □

The deadline can reduce total research while preserving the wrong service order. This is a different failure from an externally imposed time price that is simply too low. A deadline with case-specific priority reassignment is a different intervention and can work.

6.3 State-contingent opportunity-loss charge

Replace the common charge by a posted priority-right rule. If A claims the first slot, it pays $u_B = b_B^+ - w_B$ once. If it releases the slot, it pays zero. Deferred experiments and B's use are uncharged. B can decline research and act; A can retain its second-slot option. The charge is public, computed before choices from the competing case's primitives, and does not count as a social expense.

Theorem 3 (public-state implementation). The posted priority-right rule implements an optimal joint stopping and scheduling policy for every parameter configuration in the model, with payoff-equivalent tie choices allowed. If both cases have positive research value, A chooses first service exactly when $u_A > u_B$, otherwise it releases the slot. Waiting and action choices remain efficient.

Proof. A's payoff gain from claiming is $b_A - u_B$. Its payoff gain from releasing to a willing B and choosing its best waiting-or-act option is w_A. For positive b_A, b_B, the difference is $u_A - u_B$. B waits after an A claim if $w_B > 0$ and researches immediately after a release if $b_B > 0$. Thus the selected route is exactly the larger route in (4).

If $b_A \le 0$, its claim cannot improve on releasing and acting; B then researches if worthwhile. If $b_B \le 0$, $u_B = w_B = 0$. A claims if immediate research dominates its waiting option, and otherwise may take its identical-payoff research option in the final slot. If neither has positive b, both act. At equality either order has the same total value. No future charge distorts the second-slot option. □

Corollary 2 (when rival-state variation is necessary). Fix A with positive b_A and w_A. Suppose two public rival states both have b_B>0 and satisfy $u_B^L < u_A < u_B^H$. No single first-slot fee k, independent of the rival state and with uncharged deferred service, implements the efficient order in both states under the same tie rule. The opportunity-loss charge does.

Proof. For a constant first-slot fee, A compares $b_A - k$ with w_A, so it claims exactly when $k < u_A$, independently of B's state. Efficiency requires claiming in the low state and releasing in the high state, contradictory strict requirements on k. Posting u_B implements each state by Theorem 3. □

The rule charges the rival's *loss from deferral*, not its entire research value. Charging b_B^+ would overstate the externality whenever $w_B > 0$. Repeating the priority toll on a later slot would charge for the same lost option twice and could recreate underresearch. This is precisely why the distinction between a first-slot right and every unit of research is substantive.

Implementation requires observable case states, enforceable priority rights and marginal budget incentives. A recommendation that does not affect behavior is not an implementation. The rule is feasible as an internal booked priority charge or as equivalent dispatcher assignment with recorded opportunity cost. It does not require an outside payment. It is not claimed budget balanced among owners, a truthful auction, or valid under private misreporting. Dynamic pivot and congestion pricing precedents remain the appropriate conceptual comparison.

7. Recoverability separates research value from priority

Consider a case in the interior region $b_i > 0, w_i > 0$. Its priority loss simplifies to

$$u_i = h_i b_i + \ell_i = h_i s_i M_i - h_i c_i + s_i d_i. \tag{6}$$

Theorem 4 (recoverability-priority reversal). Holding the prior, signal law, cost, and expiration fixed, within this region

$$\frac{\partial b_i}{\partial r_i} = -S_i[(m_i - \bar{m}_i) + h_i \bar{m}_i] \leq 0,$$

$$\frac{\partial u_i}{\partial r_i} = h_i s_i S_i \bar{m}_i \geq 0. \tag{7}$$

The inequalities are strict when their respective coefficients are positive. Therefore increasing recoverability can lower a case's standalone research gain while raising its scheduling priority. Against a rival with fixed u_j, a priority reversal occurs whenever two recoverability levels inside this region satisfy $u_i(r_0) < u_j < u_i(r_1)$.

Proof. Differentiate g_i and M_i in (1): their derivatives are $S_i m_i$ and $S_i \bar{m}_i$. The derivative of b_i is $s_i S_i \bar{m}_i - S_i m_i$. With $w_i = s_i b_i - \ell_i > 0$, (6) follows algebraically and its derivative yields (7). Theorem 1 converts crossing losses into a reversed order. □

The economic reason is conditional. Recoverability raises the value preserved by acting in the wrong state. Within the region where research still occurs, it also raises the expected reward whose availability is threatened by expiration during an extra waiting slot. The value of improving the choice falls, but the cost of delaying that research-and-action option can rise. Once deferred research becomes unattractive, $u_i = b_i^+$ and the rising-priority conclusion need not hold. At $h_i = 0$, the positive derivative in (7) vanishes.

This reduced-form result is not a theorem that reversible trials should be prioritized. A trial that generates new information after action or changes the opportunity's hazard requires a different model. Nor is increased recoverability equivalent to a lower loss from expiration: expiration destroys the entire still-unresolved opportunity here. If salvage survives expiration, (1) and the comparative statics change. The theorem makes those boundaries explicit.

8. Correlated states: the smallest nonseparable extension

8.1 Information and the contingent joint policy

Keep the same two cases, two slots, one experiment per case, holding/expiry technology and immediate terminal-action option. Replace the product prior by a public joint binary distribution

$\mu = (\mu_{00}, \mu_{01}, \mu_{10}, \mu_{11})$. Conditional on the latent state vector, signal errors remain independent. Survival is independent of latent states and signals. Terminal action locks a payoff, but the realized action outcome is not revealed during the two-slot window; otherwise correlated cases could learn for free from action outcomes. Only research signals provide new state information here.

Let L be the live unresolved cases and E the cases with an unused experiment. Observing signal y from research on k updates the *whole* joint distribution to $\mu^{k,y}$. Write $g_i(\mu)$ for immediate-action value and

$$M_{i|k}(\mu) = E_y g_i(\mu^{k,y}), \qquad b_k(\mu) = s_k M_{k|k}(\mu) - c_k - d_k - g_k(\mu).$$

The cross-information net gain for a passive case i retained while k is researched is

$$a_{i|k}(\mu) = s_i s_k [M_{i|k}(\mu) - g_i(\mu)] - \ell_i(\mu), \quad \ell_i(\mu) = d_i + h_i g_i(\mu). \tag{8}$$

The factor $s_i s_k$ matters: i must remain actionable and k must survive to deliver the signal. If k expires, i still acts on the unchanged posterior, provided i survives. A previously researched case can now have $a_{i|k} > 0$ and should not be automatically retired.

Theorem 5 (complete correlated two-slot solution). With one slot left,

$$V_1(\mu, L, E) = \sum_{i \in L} g_i(\mu) + \max \left\{ 0, \max_{k \in L \cap E} \left[b_k(\mu) + \sum_{i \in L \setminus \{k\}} a_{i|k}(\mu)^+ \right] \right\}. \tag{9}$$

If k is selected, retain each other live case i exactly when $a_{i|k} > 0$; otherwise act on it before the signal. After survival and signal revelation, choose posterior-optimal terminal actions.

At the initial boundary choose a retained subset $R \subseteq \{A, B\}$ and research $k \in R$ or $k = \varnothing$. Define $E_k = \{A, B\} \setminus \{k\}$ when k is a case, and $E_\varnothing = \{A, B\}$. For each survivor subset $S \subseteq R$ let $P_R(S) = \prod_{i \in S} s_i \prod_{i \in R \setminus S} h_i$. Let $\mathcal{T}_k V_1$ be the signal expectation of $V_1(\mu^{k,y}, S, E_k)$ if $k \in S$, or $V_1(\mu, S, E_k)$ otherwise. Then

$$Q_0(R, k) = \sum_{i \notin R} g_i(\mu) - \sum_{i \in R} d_i - \mathbf{1}\{k \neq \varnothing\} c_k + \sum_{S \subseteq R} P_R(S) \mathcal{T}_k V_1, \tag{10}$$

$$V_2(\mu) = \max_{R,k} Q_0(R, k). \tag{11}$$

There are exactly eight root choices: stop both; keep only A with research or idling; keep only B with research or idling; keep both with A research, B research, or idling. Equations (8)-(11), involving only finite sums, posterior updates and maxima, solve every initial and signal-contingent decision; no general queue equation is left unsolved.

Proof. In the final slot either all cases act or one unused experiment is selected. Its own payoff relative to acting is b_k. For another case i, retaining it produces $-d_i + s_i[(1 - s_k)g_i + s_k M_{i|k}]$, so its incremental gain is (8). Choose retention separately by its sign; terminal payoffs are additive and there are at most two cases. Final idling gives no information and is dominated by immediate action because costs are nonnegative and rewards nonnegative. This proves (9). For the first slot, outside-R actions are irrevocable, every retained case pays its holding cost once, and the selected experiment costs c_k once. Condition on independent survival and the signal, if delivered; the remaining optimum is (9). The eight choices exhaust the initial action set, including idle policies. This proves (10)-(11) and the contingent policy. □

For a product prior, $M_{i|k} = g_i$ when i differs from k, so every passive cross gain is nonpositive. Previously researched cases then retire immediately, and (9)-(11) reduce to Theorem 1. This proves an exact separable boundary rather than treating correlation as another scalar delay cost.

8.2 Local behavior and a signed incentive boundary

The local-priority game extends the earlier one without preventing cross-learning. In the last slot, if A still has an experiment, it compares its own charged research gain $b_A - \tau$ with the gain $a^+_{A|B}$ from releasing to B, provided B has an experiment and $b_B > \tau$. If A claims, B retains only when $a_{B|A} > 0$. If A releases, B researches only when its own $b_B > \tau$, and A retains only when $a_{A|B} > 0$. If A has no experiment, B has the research choice. Each owner may act immediately rather than wait. Ties between research and action favor action; otherwise use the earlier priority convention. This explicitly solves the last-slot local game.

For completeness the root game is also finite and solved. Let $Q^L_i(R, k)$ be case i's component of (10) when its final continuation is the just-described local game rather than (9), and subtract any own research charges in computing its private component P^L_i. A initially chooses claim, release-and-retain, or act-and-release. B's feasible root choices in those three branches are respectively

$$\mathcal{C} = \{(\{A\}, A), (\{A, B\}, A)\},$$
$$\mathcal{R} = \{(\{A\}, \varnothing), (\{A, B\}, B), (\{A, B\}, \varnothing)\},$$
$$\mathcal{X} = \{(\varnothing, \varnothing), (\{B\}, B), (\{B\}, \varnothing)\}.$$

In each branch B selects a maximizer of P^L_B; A then selects the resulting branch maximizing P^L_A. This is an explicit two-level maximum over 2, 3, and 3 choices with known finite continuation values. It specifies off-path actions as well as the equilibrium. Appendix D proves the backward-induction construction and the code reports all route payoffs. It reduces to the earlier local equilibrium for independent states.

Theorem 6 (information-credit boundary). At a last-slot history with live cases i,j and only i's experiment remaining, joint research is strictly optimal if and only if $b_i + a^+_{j|i} > 0$. Local research at a nonnegative common fee requires $b_i > \tau$. Therefore, if

$$b_i \le 0 < b_i + a^+_{j|i}, \tag{12}$$

no nonnegative research charge implements the joint-optimal continuation. A public credit $\kappa_i = a^+_{j|i}$ conditional on conducting the experiment, with no further charge, implements the joint research-or-action decision. A credit strictly exceeding $-b_i$ is necessary for strict local research in the negative-own-gain case; its infimum is $-b_i$. These statements concern the last-slot decision, not a claim that this credit alone implements all root incentives.

Proof. Equation (9) compares exactly those two choices when only one experiment remains. The passive owner retains iff its own a is positive. The active owner's charged gain is $b_i - \tau$, which is nonpositive under (12). With the credit its gain becomes $b_i + a^+_{j|i}$, identical to the joint comparison. Equality can select action, which is then jointly optimal. Strict local research requires $b_i + \kappa_i > 0$. □

The independent common-fee classification and positive first-slot toll remain correct on the product-prior subspace; they do not extend by substituting marginal beliefs into b,w,u. Correlation can create both an incentive to free-ride on another case's signal and an incentive to stop socially valuable cross-informing research. A blanket assertion that local objectives always overacquire evidence would now be false.

A sufficient *full* public-state implementation is available if unrestricted signed internal transfers and observable eventual resource payoffs are allowed: give owner i its own resource payoff plus the other case's resource payoff minus a fixed initial constant. Each owner then maximizes total firm payoff minus a constant at every decision. Backward induction through the finite game implements (9)-(11). This elementary team-payoff alignment is not a new pivot theorem, not

budget balanced between owners, and not a private-information mechanism. It requires stronger incentive and accounting authority than the one-time toll. The last-slot credit alone should not be described as implementing the full root policy.

8.3 Exact positive-expiry example and contingent choices

Take $\mu = (.58, .02, .02, .38)$, so both marginal state-1 probabilities are .4 and all joint states have positive probability. Let $S_A = S_B = 100$, $r_A = r_B = 0$, $d_A = d_B = 0$, $h_A = h_B = .01$, $q_A = .8, c_A = 1$, and $q_B = .95, c_B = 30$. These are arbitrary synthetic model units.

The unique best root choice retains both and researches A first, yielding

$$V_2 = \frac{249523577}{1562500} = 159.69508928.$$

The complete root comparison is:

Retained cases	First research	Optimal joint continuation payoff
None	None	120.000000
A	None	137.418000
A	A	138.200000
B	None	123.409500
B	B	124.050000
A and B	None	153.176525
A and B	A	159.695089
A and B	B	154.405320

If both survive and A's signal is 1, the joint posterior is $(29/110, 1/110, 2/55, 38/55)$. The posterior marginals are $p_A = 8/11$ and $p_B = .7$. B's own last-slot research gain is -5.95, but A's net gain from being retained to learn B's signal is $9401/550 = 17.09272727$. Thus the joint incremental gain is $12257/1100 = 11.14272727$. Research B and retain A, even though A has already used its experiment. This is a supported branch with probability .431244 from the root.

If both survive and A's signal is 0, act on both immediately: the second experiment is not jointly worthwhile. If A expires but B survives, B's posterior is unchanged and B researches, with own gain 4.05. If only A survives, it acts on its observed posterior. If neither survives, there is no action reward. These six branches are the complete contingent optimum and are reported with exact probabilities in the reproduction output.

In the solved local game A instead releases to B initially to obtain more valuable free cross-information; the total payoff is 154.40532. The R1 priority toll calculated only from initial marginals is .6405 and gives the same suboptimal outcome. Independently of this root comparison, every nonnegative uniform research fee fails to implement the efficient full policy, since the positive-probability second-slot branch requires B to research with negative own gain. The unpriced common-deadline payoffs are 120 for zero slots, 154.22514 for one slot, and 154.40532 for two slots, all below the joint optimum. Unlike the independent proof, these comparisons use the correlated local game.

For an exact independence check, replacing the joint prior by $(.36, .24, .24, .16)$ preserves both marginals and every other primitive. The resulting joint and local values both equal 141.6095 and the passive cross-information gain is nonpositive. For a simple boundary check with perfect positive correlation and no expiry, $\mu = (.6, 0, 0, .4)$ gives a joint value 165.4 versus local 160. These are three exact small instances, not a simulation campaign or empirical estimates.

The rational solver and independent physical-outcome enumeration agree on all eight root values in each instance. The enumerator includes contingent retention of the already researched

case, actual terminal action success/error, and all positive-probability signal and expiry paths. There are 63 policy maps in each full-support instance and 30 in the zero-expiry boundary, 156 total. Final idle policies are analytically dominated; initial idle policies are enumerated. The new script reports 37 checks. A separately evaluated full physical tree, not merely the same Bellman recursion twice, supplies the comparison.

8.4 Two public-history credits: complete implementation

The final-slot information credit in Theorem 6 does not resolve A's initial incentive to free-ride on B's information. We now close that gap within the **same** model using only two nonnegative internal credits. Credit C is paid to A when its initial-slot experiment is conducted, even if A then expires. Credit κ is paid to B when its final-slot experiment is conducted after initial research on A, both cases surviving that first slot, and public signal $Y_A = 1$. Payment precedes second-slot survival. All other payments are zero, including every research fee. The announced credits are committed and enter the owners' quasilinear objectives; they depend on public research/history, never eventual action rewards. They do not add to firm resource losses.

Keep the synthetic full-support witness's primitives: $p_A = p_B = .4$, $q_A = .8$, $q_B = .95$, $c_A = 1$, $c_B = 30$, $h_A = h_B = .01$, $S_A = S_B = 100$, and $d_i = r_i = 0$. Vary only

$$\mu(\delta) = (.36 + \delta, .24 - \delta, .24 - \delta, .16 + \delta), \qquad -.16 \le \delta \le .24.$$

We prove the following result on the exact, nonempty closed interval

$$I = [17/100, 23/100]. \tag{13}$$

This is a certified interval around .22, not an asserted maximal region or a classification of the entire admissible prior range. Independence remains at $\delta = 0$, outside I, and retains Theorems 1-4's appropriate scope.

The efficient contingent policy π^* on I retains both cases and researches A initially. If both survive and $Y_A = 1$, it retains A and researches B; if $Y_A = 0$, both act immediately. If A expires and B survives, B researches without a credit. If only A survives, A acts on its own signal. Neither survivor means no terminal reward. After the second experiment, survivors act using exactly their available public signals. With $s = 99/100$, its resource components are

$$X_A(\delta) = \frac{43226820450\delta + 33275667541}{500000000},$$
$$X_B(\delta) = \frac{297(495000\delta + 515039)}{2500000}. \tag{14}$$

Define the credit-event probability, two switch points, and B's continuation deficit by

$$p = \frac{107811}{250000} = s^2 \Pr(Y_A = 1), \quad \delta_0 = \frac{7831}{45000}, \quad \delta_1 = \frac{5291}{30000}, \quad K(\delta) = \frac{30000\delta - 5291}{220}. \tag{15}$$

Here p is **not** multiplied by second-slot survival: B is paid on conducting research. Define A's required initial compensation

$$F(\delta) = \begin{cases} (877679550\delta + 171175559)/500000000, & \delta \le \delta_0, \\ (449821795 50\delta - 7503987541)/500000000, & \delta \ge \delta_0. \end{cases} \tag{16}$$

The pieces agree at δ_0 and F is strictly positive throughout I.

Theorem 7 (two-credit implementation and initial toll failure). On I the root action and supported continuation of π^* are strictly efficient. Under the stipulated zero-gain action tie and positive initial-priority tie conventions, the two-credit local game implements its entire contingent policy if and only if

$$C \geq F(\delta), \qquad \kappa \geq 0, \qquad \kappa > K(\delta). \tag{17}$$

Strict implementation at the initial comparison and every supported discretionary research/retention decision additionally requires $C > F(\delta)$. A positive tie between initial claim and release selects A's initial claim; at $\kappa = K \geq 0$, B instead selects action and implementation fails. The strict-implementation statement concerns the efficient outcome and its supported decisions, not elimination of all possible indifferences in off-path subgames.

Replacing C by an initial A-claim **toll** $T \geq 0$, while retaining the same B-only continuation credit and zero other payments, cannot implement π^* anywhere on I. This is a claim about that restricted instrument class, not arbitrary taxes, subsidies, or budget-balanced mechanisms.

Proof. The joint-policy certificate and all off-path continuation calculations are given in Appendix E. Conditional on both cases alive after $Y_A = 1$, B's own research gain is $-K$, and A's net passive cross gain is

$$a_{A|B} = \frac{4410450\delta - 594259}{22000} > 0.$$

Their joint increment is $(1410450\delta - 65159)/22000 > 0$ on I. B conducts research exactly when $\kappa > K$. A then strictly retains. After $Y_A = 0$, B's uncredited research gain is $-3(5000\delta - 189)/140 < 0$; neither research nor idle continuation is chosen. On the B-only survival branch, its uncredited gain is $81/20 > 0$. All these decisions use the actual local continuation, not a prescribed dispatcher.

When $\kappa > K$, B's root private payoff from retention after an A claim is $X_B + p\kappa > 60$, so it strictly retains instead of acting. If A releases and remains unresolved, B strictly researches initially: its own value is $1281/20 = 64.05$, versus $126819/2000 = 63.4095$ from waiting and 60 from acting. These comparisons include the off-path no-signal final game. The same choices hold if A acts and releases. A's release-and-retain resource payoff is

$$R_A(\delta) = 51.54336 + 176.418\delta + \frac{401841}{1000000}\left[\frac{7831 - 45000\delta}{205}\right]^+. \tag{18}$$

This includes A's own later research when beneficial after B's first signal, and research if B expires without delivering a signal. Direct subtraction gives $R_A - X_A = F > 0$. A's three resulting private payoffs are $X_A + C$, R_A, and 60. Since $R_A > 60$, the initial comparison is exactly $C \geq F$, with strictness as stated. If $\kappa \leq K$, a positive-probability efficient continuation is absent regardless of C; necessity follows. Conversely all the displayed strict continuation comparisons and the root comparison imply the complete target policy, proving sufficiency. With a toll the claim payoff is $X_A - T < R_A$ whenever B's target continuation is induced; if not induced the continuation already fails. This proves the restricted toll impossibility. □

The threshold at $\delta_0 \simeq .17402222$ comes from A's uncredited own experiment after B's initial signal 1, an off-path continuation that changes its release payoff. The threshold at $\delta_1 \simeq .17636667$ comes from B's credited on-path experiment changing from privately profitable to nonpositive. Both are inside I. The full game therefore yields piecewise initial and continuation requirements even though its efficient policy is unchanged.

8.5 Funding frontiers, attainment, and exact witness

Let $k(\delta) = \max\{0, K(\delta)\}$. A gross expected-credit constraint $\bar{E}$ limits expected transfers, while gross pathwise capacity $\bar{H}$ must fund the largest possible payment total. These are institutional limits on credits, not welfare-cost terms. Under implementation, A is always paid C and B is paid κ on an event of probability p. Therefore

$$E = C + p\kappa, \qquad H = C + \kappa. \tag{19}$$

Theorem 8 (restricted funding frontier). For weak implementation selected by the stated ties, the two infima are

$$E_{\inf}(\delta) = F(\delta) + pk(\delta), \qquad H_{\inf}(\delta) = F(\delta) + k(\delta). \tag{20}$$

If $\delta < \delta_1$, both are attained together at $(C, \kappa) = (F, 0)$. If $\delta \geq \delta_1$, neither is attained. Strict supported implementation has the same two infima and attains neither at any $\delta \in I$.

For a weak implementation, the exact budget-feasibility conditions are

Prior region	Expected cap only	Pathwise cap only
$\delta < \delta_1$	$\bar{E} \geq F$	$\bar{H} \geq F$
$\delta \geq \delta_1$	$\bar{E} > F + pk$	$\bar{H} > F + k$

For strict supported implementation, both respective conditions are strict throughout I. If both budgets are imposed, **both** corresponding inequalities are necessary and sufficient. Expected and pathwise caps cannot be substituted for each other. These statements completely classify funding inside the two-credit class, without optimizing over other contracts.

Proof. Equation (17) is the complete feasible payment set. Both objectives in (19) increase strictly in each payment since $0 < p < 1$. When K is negative, $(F, 0)$ is feasible and minimizes both. At K zero or positive, B's strict research incentive requires $\kappa > k$; approach but do not reach k while C equals F. Strict root implementation also requires C above F. These facts prove the infima and attainment statements. For simultaneous caps with positive slack, choose sufficiently small nonnegative increments above F and k; the expected increment is $\epsilon_C + p\epsilon_\kappa$, and the pathwise increment is $\epsilon_C + \epsilon_\kappa$. They fit both caps. If K is negative and only weak implementation is required, the attained pair fits both weak caps directly. Necessity follows from monotonicity. □

At the preserved witness $\delta = 11/50$, the exact values are

$$\begin{aligned} X_A = \frac{1069639201}{12500000} = 85.57113608, \quad R_A = \frac{2258883}{25000} = 90.35532, \\ F = \frac{59802299}{12500000} = 4.78418392, \qquad K = \frac{119}{20} = 5.95, \\ E_{\inf} = \frac{183752143}{25000000} = 7.35008572, \quad H_{\inf} = \frac{134177299}{12500000} = 10.73418392. \end{aligned} \tag{21}$$

Neither weak nor strict implementation attains these witness infima: B chooses action at $\kappa = 5.95$. At that limiting value B's root retained-route private value is $X_B + pK = 76.689855 > 60$, verifying that retention is slack. For example $C = F + .01$ and $\kappa = K + .01$ implement strictly, with $E = 7.36439816$ and $H = 10.75418392$. The firm's physical payoff remains 159.69508928; adding or subtracting the credits from it would conflate internal funding with resource accounting.

The exact endpoint certificate and witness computations enumerate physical latent states, signals and survival with rational arithmetic, independently of the joint-value solver. They

verify 63 contingent schedule/retention maps at each of two interval endpoints and the retained witness. This is a proof certificate and accounting check, not a parameter grid or new simulation campaign. The 101 checks include all root alternatives, complete credited local-game choices, both binding incentive endpoints, piece continuity, and the original fractions.

9. Exact synthetic verification

9.1 Independent computation and counting

All reported numbers are synthetic model units. The retained independent-case script `joint_analysis.py` computes (1)-(5) and the posted policy. Independently, a second evaluator enumerates every root choice of retained subset and active case or idling, every feasible signal-contingent final-slot research map, both latent states, both signal outcomes, and all positive-probability survival paths. It values realized correct and incorrect terminal actions and actual costs directly; it does not use the b, w, u formulas in the evaluator.

With strictly interior hazards the full two-case tree contains 35 distinct scheduling policies in this representation: immediate action on both; one-case research; one-case idling with terminal choice; two-case idling with survivor-contingent scheduling; and research on either case with signal-contingent final allocation. When zero-probability branches are removed the count can be smaller. Terminal actions after each available signal use Bayes-optimal classification; all possible research schedules are enumerated. The separate posterior computation does not import the closed-form scheduling optimum.

For the six named scenarios the output contains 193 policy rows, including 35 for each interior-hazard example and 18 when both hazards vanish. The script also checks additional parameter configurations and the policy boundaries; 250 independent-case checks pass. These checks validate implementation in this finite model, not external validity or originality. The earlier single-case script separately passes 12,002 checks and remains in the reproduction package.

9.2 Congestion, charge failure, and deadline failure

Let both cases have $p = .5, S = 100, r = 0, c = 1, d = 0$. Let A have $q_A = .7, h_A = .02$, and B have $q_B = .9, h_B = .20$. Then

$$b_A = 17.6, \quad w_A = 16.248, \quad u_A = 1.352,$$
$$b_B = 21, \quad w_B = 6.8, \quad u_B = 14.2.$$

The joint optimum researches B first, retains A, and obtains $100 + 21 + 16.248 = 137.248$. Local priority gives A first and B later, yielding 124.4. A one-slot common deadline yields 117.6; an immediate deadline yields 100. For every common charge $\tau < 17.6$, A still takes the first slot; for $\tau \geq 17.6$, A does not perform the valuable deferred experiment. Thus no common charge implements the optimum, a conclusion from the theorem rather than a finite tax grid. The posted priority charge is 14.2. A compares $17.6 - 14.2 = 3.4$ with its retained value 16.248 and releases the slot, implementing the optimum.

Changing only B's prior to .705 in this example gives $g_B = 70.5$, $b_B = .5$, $w_B = 0$, and $u_B = .5$. A should then be first, with total payoff 138.1. A's own primitives are unchanged. The pair of B priors .5 and .705 straddles A's priority loss, so even a constant first-slot fee cannot implement both states. This demonstrates why the correction must respond to the competing belief rather than merely to the focal analyst's own research cost.

9.3 Priority reversal and positive implementation examples

For the reversal, keep A as above and set B to $q_B = .7, h_B = .022$, with the other primitives unchanged. Initially $u_A = 1.352 < u_B = 1.48412$, so B is first. Increase A's recoverable share to .5. Its research gain falls from 17.6 to 7.3, but its priority loss rises to 1.646; both research-retention inequalities remain strict. A is now first. The payoff levels differ because recoverability changes the primitive rewards; the comparison identifies an ordering reversal, not a treatment-effect estimate.

If both hazards and holding costs are zero, $w_i = b_i^+$ and all deferral losses vanish. Both positive-value experiments can be performed in either order at no loss, and a zero common charge works. If A's acquisition cost rises to 30 in the congestion example, $b_A = -11.4$ and it acts immediately; B researches first at zero common charge, also efficiently. These examples prevent an unconditional claim that constant charges or existing priority rights always fail.

10. Welfare measurement and prospective queue experiment

The proposed empirical unit changes with the shared-resource mechanism. The primary unit is a pre-registered analyst-queue episode containing a frozen cohort of eligible decisions and a specified capacity window, not a completed individual case. The estimand is the intention-to-treat effect of an offered opportunity-loss scheduling aid on total payoff over all cases in that episode. Per-case effects and cross-case spillovers are secondary. No firm data, partner agreement, recruitment, treatment assignments, or implemented experiment exist.

The protocol crosses an incremental evidence packet with a queue-level priority aid, keeping analyst capacity and the economic alternatives comparable. The aid explicitly records immediate action, deferred research, expiration, and the foregone value of the competing case. It compares a local-order baseline with opportunity-cost scheduling. It does not purport to identify the behavior of real budget charges if only a recommendation is offered. Any charge experiment would require actual internal authority and an additional registered arm. A common-deadline arm is neither necessary nor currently authorized. If cases have correlated underlying conditions, record cross-case prior links, posterior revisions after each public signal, and decisions to retain an already researched case. The independent u-score is not applied to such cases as if a marginal-prior update were sufficient; the two-slot correlated formula or an explicitly limited recommendation must be used.

Randomization must occur at the entire shared analyst pool or independent pool-period level, with buffer periods and all displaced work tracked. Randomizing individual cases within the same queue would expose controls to treatment-induced delay and cannot identify an isolated case effect without an interference design. The episode cohort includes rejected, expired, and unresolved cases. Primary payoffs include every case and every worker, not just those served before the measurement window ends.

Double counting is a central risk. Actual action margin, research cost, residual holding cost, expiration consequence, and correction/rework expense must have exclusive ledger categories. The theoretical u_j already summarizes the rival's modeled forgone option and is not added to its observed payoff loss. Internal tolls are transfers. Uncertain costs produce conditional valuation bounds rather than arbitrary monetary estimates. Confidence intervals must account for both assignment and partial identification.

If only d_i is interval valued, every comparison remains conditional on priors, signal laws, terminal rewards, expiration, and feasible actions. If those are also uncertain, the entire set of candidate joint policy values must be recomputed over a documented joint uncertainty set. A policy is certified only if its worst comparison is nonnegative throughout that set. No unconditional model-robust claim follows from bounding delay prices alone.

The detailed prospective protocol contains queue and case dictionaries, inclusion rules, interference assumptions, missing-outcome bounds, and illustrative precision calculations. Public BPI Challenge 2019 documentation provides procurement-process context; its repository file and license were not verified and no rows were analyzed. It cannot identify the proposed mechanism without resource rosters, evidence requests, complete intake, and linked economic outcomes.

11. Conclusion

The scarce resource is research, not terminal authorization; action consumes no shared capacity. There are exactly two cases, one experiment per case, and two slots. Sections 4-7 assume independent states; Section 8 solves the explicitly correlated counterpart. Neither model includes more experiments per case, arrivals, heterogeneous durations, endogenous effort, private reporting, or commitment problems. Terminal outcomes do not reveal latent states during the two-slot decision window; only public research signals provide new state information. We do not present an unsolved general queue as a completed result.

The bounded independent-case contribution is the recoverability/priority derivative and the exact common-charge implementation set, not the scheduling interchange or public externality charge themselves. Under correlation, the first signal changes the second case's research and retention values. A supported branch can require research despite its negative own-case gain, so the positive-toll-only extension fails. The full contingent joint policy and finite local game are solved. Theorems 7-8 close the initial incentive gap for the two specified credits, with exact piecewise thresholds and expected/pathwise funding limits. Their nonnegative-payment constraint and public-history support are essential scope conditions.

These results are stronger than a separable benefit/time ratio but remain applications of established scheduling and externality principles. Bergemann-Välimäki provides a much broader mechanism-design framework, and expiring/restless scheduling is a substantial existing literature. The strongest remaining objection is that the small public-state implementation characterization and comparative static may still be too modest for a leading general-interest journal. No amount of synthetic verification establishes empirical relevance or priority. The two-credit frontier is a complete restricted implementation result, but uses standard incentive comparisons and Bayesian backward induction and does not certify a substantial originality claim. It does not solve private reports, optimal choice of payment support, budget balance, endogenous funding, or arbitrary correlations outside the certified interval. No empirical evidence resolves practical importance. This is an integrated revised theoretical paper with a separate prospective study, not a claim of scientific acceptance.

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Appendix A. Shared-resource model: complete derivations and verification

A.1 Binary signal and posterior actions

For case i with prior p and accuracy q, the joint probabilities of state 1 and signal 1, state 0 and signal 1 are pq and (1-p)(1-q). The joint probabilities for signal 0 are p(1-q) and (1-p)q. Bayes-optimal expected classification accuracy is

$$a(p,q) = \max\{pq, (1-p)(1-q)\} + \max\{p(1-q), (1-p)q\}.$$

Thus $\bar{m} = 1 - a(p,q)$ and $M = S[r + (1-r)a(p,q)]$. At any zero-probability signal branch, the posterior action is irrelevant to the expected payoff. At a posterior tie either terminal action has the same value. The enumeration uses a fixed state-1 tie convention. The formula applies to asymmetric priors without presuming that the signal always overturns the prior-optimal action.

A.2 Deferred value and no double counting

If the case acts now, it receives g. If it waits one slot before deciding whether to research, it incurs d immediately; expiration destroys the opportunity with probability h. Conditional on survival the last-slot optimal value is $g + b^+$. Waiting's incremental payoff is

$$-d + s(g + b^+) - g = -\ell + sb^+.$$

The best of action now and this option gives $[-\ell + sb^+]^+$. Since $\ell \geq 0$, this equals $[sb - \ell]^+$ for every real b, which is w. In particular, w>0 implies b>0. The expression includes the cost of the later experiment through b, the first waiting slot's cost through ell, and two distinct opportunities for expiration through s and b. It does not charge a separate hypothetical missed opportunity in addition to those losses.

For i first and j waiting, a direct physical-outcome calculation gives

$$s_i M_i - c_i - d_i - d_j + s_j(s_j M_j - c_j - d_j).$$

Subtract $g_i + g_j$ and rearrange to $b_i + s_j b_j - \ell_j$. If waiting j is not profitable, immediately act on j and obtain $G + b_i$. Taking their maximum yields $G + b_i + w_j$. This identity is checked by the independent latent-outcome evaluator, not assumed in that evaluator.

A.3 Exhaustiveness of Theorem 1

At the initial boundary there are four retained subsets, including the empty one. For each retained subset R, the active choice is one case in R or no research. Cases outside R act immediately. If some case is active, its experiment uses its only research opportunity; if it survives, action immediately after its signal weakly dominates further waiting because no additional own-state information or reward becomes available. The independent state assumption means its public signal cannot improve the remaining case's posterior. Therefore the only remaining decision is whether an unserved case was retained and, if it survived, whether to research it in the last slot. This gives the two research-first candidates in Theorem 1, with optional immediate retirement of the unserved case embedded in w.

If no case is active initially, fix the retained subset R. Its expected value relative to immediate action on all cases is

$$-\sum_{i \in R} \ell_i + E \max\{0, \{b_i : i \in R \text{ survives}\}\}.$$

The expectation is bounded above by $\max(0, b_A, b_B)$. Immediate research on a positive maximizer, if one exists, followed by immediate action on the other case attains that upper bound without incurring the idle waiting losses. If none exists, stopping both dominates idling. This proves exhaustion even for policies that idle while retaining only one case. Signal-dependent final research choices cannot improve on w, because the other case's independent signal changes neither b nor the feasible final choices.

For both b positive, order comparison is exactly

$$[b_A + w_B] - [b_B + w_A] = u_A - u_B.$$

When b is nonpositive, that case's w is zero and it never needs research; Theorem 1's maximum also captures these corners. A case with h=0,d=0 can be deferred without loss. This yields payoff ties, not a failure of the formula.

A.4 Sequential equilibrium with a common charge

The game is complete information. At the final slot each surviving case researches only if its charged gain b-tau is strictly positive. At the initial boundary A moves first. If A researches now, B's best wait gain is $[s_B(b_B - \tau) - \ell_B]^+$ and its alternative is immediate action. If A releases, B researches immediately whenever $b_B - \tau > 0$; otherwise it acts. In the latter event A cannot gain by idling instead of taking its own available experiment immediately. If A releases to a researching B, its best waiting gain is $[s_A(b_A - \tau) - \ell_A]^+$.

For b_A-tau>0, its own immediate gain minus the untruncated deferred gain is

$$h_A(b_A - \tau) + \ell_A \geq 0.$$

Truncating a negative deferred gain at zero preserves weak preference for immediate research. It is strict when ell_A>0. If both h_A and ell_A are zero, early and late gains tie and the stated early-service convention selects the first slot. For b_A-tau<=0, immediate action weakly dominates both research and waiting; the research-action tie convention selects action. Backward induction therefore yields Theorem 2, including off-path choices.

The actual firm payoff for b_A>tau is

$$G + b_A + \mathbf{1}\{s_B(b_B - \tau) > \ell_B\}\, w_B.$$

The charge affects the private choice but is not subtracted again from social payoff. For b_A<=tau it is $G + \mathbf{1}\{b_B > \tau\} b_B$. These formulas permit evaluation of all tau, not just sampled values.

A.5 Full implementation classification and tie boundaries

Assume first that the efficient first research assignment is unique. If A first is optimal, b_A>0 and tau=0 reproduces its efficient retention decision for B. If B first is optimal and b_A<=0, tau=0 implements it. If neither case has positive b, acting immediately is efficient and tau=0 implements it.

The remaining case has unique B first and b_A>0. If w_A=0, strict optimality implies b_B>b_A+w_B>=b_A. A releases and stops for tau>=b_A, while B researches for tau<b_B; hence [b_A,b_B) is sufficient and, under the early-service convention, necessary. If w_A>0, tau<b_A preserves A's first claim and tau>=b_A destroys its later research incentive. Thus no tau works under that convention. With ell_A>0 this conclusion is independent of first-slot tie-breaking because A strictly prefers early service for every tau<b_A.

If ell_A=0, h_A=d_A=0 because g_A>0. For tau<b_A A can be indifferent between early and late service. Selecting release in that tie can support the efficient B-first route. The stronger claim that *no equilibrium with any tie-breaking* implements the optimum would therefore be false. We do not make it. At ties in the joint optimum, implementation requires attaining any joint-optimal policy, not a predetermined labeled ordering. The formulas in the main text directly test such cases.

A common deadline can leave zero, one, or two slots because the model is discrete. Equation (5) is the exhaustive equilibrium payoff list for an unpriced unchanged priority game. A strict B-first optimum with b_A>0 exceeds all three. If deadline design includes changing priority rights, case-specific deadlines, or contingent exceptions, it is outside this impossibility statement.

A.6 Opportunity-loss pricing and transfers

Corollary 2 follows because a fixed first-slot fee k induces the same comparison $b_A - k$ versus w_A in every rival state. With A fixed and both rivals willing to use a released slot, low and high rival deferral losses that straddle u_A require opposite strict inequalities on k. In the synthetic belief example, changing only B's prior from .5 to .705 changes its loss from 14.2 to .5, straddling u_A=1.352.

The posted first-slot charge for A is $u_B = b_B^+ - w_B$. If both cases have positive b, A compares its early gain $b_A - u_B$ with its release-and-retain gain w_A. The difference is u_A-u_B. B's first-service and deferred-service choices are uncharged, so b_B and w_B are undistorted. The resulting joint policy selects exactly a maximizing route in Theorem 1. The same argument applies if A immediately acts after release because w_A=0.

If b_B<=0, u_B=w_B=0, B immediately acts. A's early versus deferred comparison is b_A versus w_A. If these tie at a positive value, A has h_A=d_A=0, so waiting and using the otherwise idle final slot is payoff equivalent. If b_A<=0 it cannot obtain positive gain by claiming or waiting. Both nonpositive cases stop. This proves all corners of Theorem 3.

The charge equals the change in B's optimal incremental payoff between being offered first service and being offered deferred service. It is not B's gross reward or all of its evidence value. A central budget can retain it. The implementation claim assumes marginal booked transfers enter objectives and cannot be evaded; it does not establish budget balance, individual participation relative to every alternative institutional entitlement, or private-type revelation. The general dynamic-pivot literature is therefore a precedent, not a theorem proved anew here.

A.7 Comparative statics and a reversed order

Let delta=m-mbar>=0. At fixed signal law,

$$\partial_r b = sS\bar{m} - Sm = -S(\delta + h\bar{m}).$$

Within b>0,w>0,

$$u = b - (sb - \ell) = hb + \ell = hsM - hc + sd,$$

so $\partial_r u = hsS\bar{m}$. At p=.5,q=.7,S=100,c=1,d=0,h=.02, b(r)=17.6-20.6r and u(r)=1.352+.588r on this interior segment. At r=.5, b=7.3,w=5.654,u=1.646. A fixed rival u=1.48412 is crossed at r approximately .22469, inside the region where deferred research remains profitable. The exact crossing is (1.48412-1.352)/.588. All reported endpoints and ordering are in the generated output.

For other parameters within this region, $\partial_S u = hs[1 - (1 - r)\bar{m}] > 0$ when h>0, whereas $\partial_S b = s[1 - (1 - r)\bar{m}] - [1 - (1 - r)m]$ may have either sign. Increasing signal informativeness

lowers mbar and raises M; its effect on b and, in this interior, u is nonnegative. The hazard derivative of u is $(1-2h)M-c-d$, which need not be positive. Near a stopping boundary, positive-part kinks require evaluating the piecewise formula, not extrapolating derivatives. These statements do not extend without modification to trials, state-dependent expiration, correlated cases, or multiple remaining signals.

A.8 Independent enumeration: policies and physical paths

The enumerator and closed-form solver share primitive inputs only. It enumerates theta_A,theta_B; one potential binary signal for each case; and two potential survival coins per case. Unused draws integrate out. Actual action success or error is evaluated against the latent state; costs are charged only when a case is retained or researched. A case stopped at intake earns its actual prior-optimal action payoff on that latent path, not zero. A case expiring before action receives zero terminal reward.

For strictly interior hazards, the eight root actions are: stop both; retain A with active A or no research; retain B with active B or no research; retain both with A active, B active, or no research. Their feasible final-slot contingent policies number 1,1,2,1,2,8,8,12, respectively, totaling 35. For root A-active with both retained, A has three public outcomes (expiry, signal 0, signal 1); for each, if B survives the policy can act or research B. Thus there are 2ˆ3=8 maps, even though independence ultimately makes their optimal choices the same across A's outcomes. For root idle-both, the both-survive node allows three final assignments and each one-survive node two; the neither-survive node has one, giving 3*2*2=12. Zero-probability nodes are removed, reducing counts in boundary cases.

All positive-probability latent/signal/hazard paths are integrated for each map. The maximum is compared with Theorem 1. The code also compares the posted rule with the optimum over a deterministic parameter grid, verifies local payoff does not exceed it, and checks the charge-failure and recoverability examples. New validation output reports 250 checks; these are computational tests, not additional formal results or estimated observations. The primitive grid is deliberately heterogeneous and is not a calibration. Complete named-scenario policy rows are supplied in CSV, including dominated idle policies.

A.9 Conditional valuation bounds

For an implemented fixed joint policy z, expected total payoff can be written $A_z - \sum_i d_i T_{iz}$, where A includes terminal rewards and acquisition expenses, and T_i is expected unresolved case-period exposure. Expiration is already inside A and T. Holding intervals $d_i \in [l_i, u_i]$ give endpoint bounds for each fixed-policy contrast by selecting endpoints according to the signs of the exposure differences. Do not add u_i as another resource expense.

If priors, precision, stakes, recoverability, or hazards are uncertain, these quantities and even the optimal policy change. A cost-only endpoint calculation holds them fixed. For a full uncertainty set Omega, a proposed fixed policy is certified over a comparator set only if the minimum of each payoff contrast over Omega is nonnegative. Without a justified Omega and actual parameter restrictions there is no claim of model robustness. The paper implements synthetic point-primitives only.

Appendix B. Retained separable benchmark and previous checks

B.1 Existence, admissibility, and the cost frontier

The finite history tree has finitely many actions at each node. A deterministic strategy assigns stop-and-act or an experiment to each live nonterminal node. Backward induction gives an

optimal deterministic strategy for every cost price b. Randomized strategies are mixtures and cannot exceed the best deterministic expected payoff. All rewards and costs are bounded because the tree is finite.

Fix a live history x and a deterministic strategy π constrained to continue first. Let τ count the number of research choices before terminal action or expiration. Expected resource expense is $C_\pi = E_\pi \sum_{s<\tau} c_s$. Delay exposure is $T_\pi = E_\pi \tau \geq 1$. Let R_π count the action reward only on histories where action precedes expiration. Then

$$J_b(\pi) - g(x) = R_\pi - C_\pi - g(x) - bT_\pi.$$

The maximum over Π^+ is strictly positive if and only if b is below the maximum ratio. A negative ratio means that strategy is inferior even with free additive delay. When b equals the largest ratio, stopping ties with a best continuation and is selected by convention. At histories where continuation is impossible use $D = -\infty$. The binary continuation region is nested in b, even though the optimal experiment need not remain the same. This proof does not rely on a Gaussian posterior, stationary boundaries, or a one-signal approximation.

B.2 Bellman regret identity

For a continuation action e let

$$Q_d(x, e) = -c_e - d + (1 - h_e) E_e V_d(x').$$

For a terminal action a, let $Q_d(x, a) = E[u(a, \theta, x) \mid x]$. Optimal action payoff g is the maximum over terminal actions. Extend V by zero to the absorbing expired state. At a live continuation state, the realized period reward is $-c_e - d$. Its conditional expectation plus the next-state value equals Q_d. At a terminal state, realized terminal reward has conditional expectation $Q_d(x, a)$ and next value zero.

Subtract the conditional equality from $V_d(x)$, sum along the realized finite path, and apply iterated expectations. Intermediate values cancel. The initial value minus all expected realized rewards is the expected sum of $V_d - Q_d$. This yields the formula in separable benchmark. There is no survival conditioning or exclusion of incomplete cases. A deadline alters μ at every history where its anticipation changes behavior, and the identity still holds.

B.3 State-specific stopping interventions

Let μ be the original analyst policy evaluated at firm costs. At every selected history x with $V_d(x) = g(x)$, $J_d(\mu \mid x) \leq g(x)$. Changing the policy to stop there gives at least its old continuation value. At unselected histories retain the old action. By induction from terminal histories, improvements in successor continuation values weakly improve predecessor payoffs, while selected histories weakly improve because no feasible continuation beats g. Thus expected firm payoff cannot fall. If some selected reachable history had a strictly negative old continuation gain and is reached with positive probability under the new prefix, the improvement is strict. This argument presumes fixed signal technology and no queue or effort response.

B.4 Worked nonmyopic example

Let current probability of state 1 be .7, signal accuracy .7, no expiration, correct action reward 100, wrong action zero. One positive signal has probability .58 and posterior .49/.58; a negative signal has probability .42 and posterior .5. Expected maximum posterior is $.49 + .21 = .70$. Immediate one-step value of information is zero.

Two further signals yield unconditional state-likelihood pairs: positive-positive $(.343, .027)$; each mixed path $(.147, .063)$; negative-negative $(.063, .147)$. Summing maxima gives $.343 + 2(.147) + .147 = .784$. Two samples cost two when $c = 1, d = 0$, so a feasible continuation gains 6.4. This is a lower bound on optimal adaptive continuation value, sufficient to disprove the myopic converse. The delivered code checks the posterior arithmetic independently of the Bellman recursion.

B.5 Final-period derivation and boundaries

Expected payoff after one experiment followed by action is $(1-h)(S - L\bar{m}) - c - d$. Subtract $S - Lm$. Rearrangement yields $(1-h)L(m-\bar{m}) - h(S - Lm) - c - d$. Blackwell precision lowers expected Bayes classification error at fixed prior and payoff. Hazard lowers the value through both foregone learning and destruction of immediate opportunity value. The derivative with respect to r is nonpositive even though wrong-action salvage increases the level of immediate reward.

Higher stakes do not universally favor waiting. For example, at prior .5, $r = 0, q = .55, h = .2$, the coefficient of S is $.8(.55) - .5 = -.06$. With nonnegative costs, continuation is inferior at every positive stake. Here greater stakes increase the value destroyed by expiration faster than the value of learning. This is consistent with the reported high-hazard scenario, but is a separate final-period analytical example.

B.6 Delay-cost uncertainty and policy ranking

For fixed policies $z = 0, 1$, define $A_z = R_z - C_z$, so $W_z(d) = A_z - dT_z$. For $d \in [d_L, d_U]$, the contrast interval is

$$[\min\{\Delta A - d_L \Delta T, \Delta A - d_U \Delta T\}, \max\{\Delta A - d_L \Delta T, \Delta A - d_U \Delta T\}].$$

If $\Delta T = 0$, delay price cannot change the ranking. Otherwise the break-even price is $\Delta A / \Delta T$, which need not lie in the admissible interval. This formula values fixed implemented policies. It is distinct from optimizing V_d separately for every price.

With heterogeneous $d_i \in [l_i, u_i]$, use case-level extrema of $A_i - d_i T_i$. For a treatment-control difference, subtract the control upper bound from the treatment lower bound for the lower contrast, and conversely for the upper. This is conservative when cross-case cost restrictions are ignored. It does not identify a missing payoff component merely because its interval is assigned a number.

B.7 Optional stopping and inference

The model is a Bayesian economic stopping problem, not repeated null-hypothesis testing. No signal is a sequence of unadjusted p-values. The proposed trial uses a fixed final analysis date and fixed enrollment after feasibility. Exploratory interim monitoring cannot trigger an ordinary fixed-sample significance claim. If operational needs require sequential confirmatory monitoring, an alpha-spending plan or an always-valid method and its assumptions must be chosen and registered before treatment outcomes are examined. No sequential frequentist method is implemented in this package.

B.8 Computational verification

Python 3 standard-library code implements exact posterior updating on (n, k), backward induction under perceived costs, and separate forward evaluation at full costs. In the pre-R1 separable implementation, 12,002 assertions/checks pass. The parameter-grid count is 324: three

stakes, three recovery fractions, three precisions, four delay prices, and three hazards. Each compares all 36 nonterminal histories of an eight-period tree and the initial welfare ordering, yielding 11,988 checks. Seven frontier comparisons, three posterior checks, and four derivatives give 12,002 total. The posterior pair calculation is one of the three posterior checks.

For an independent check, the script enumerates all distinct pairs $(R - C, T)$ at horizon three, allowing independent policy choices at the two signal descendants. It retains 15 distinct pairs at the root and verifies their upper envelope against the Bellman solution. This is numerical agreement with a finite tree, not a proof for an infinite horizon. The policy CSV uses a 12-period horizon. Floating-point comparisons use tolerances; no random seed is required.

Appendix C. Earlier single-case synthetic illustrations

These are retained separable benchmarks, not evidence for the shared-resource mechanism. The prior is .5, signal accuracy .7, correct payoff 100, wrong payoff zero, acquisition cost 1, additive delay cost 5, expiration probability .02 per research period, and a 12-sample finite horizon. The silo internalizes ten percent of additive delay cost. Values are arbitrary model units, computed by exact backward induction and separate full-cost policy evaluation.

Policy or environment	Payoff	Periods	Complete	Accuracy
Firm optimum	62.600	1.000	.980	.700
Silo analyst	59.002	3.294	.934	.843
Silo, immediate deadline	50.000	0.000	1.000	.500
Silo, one-sample deadline	62.600	1.000	.980	.700
Silo, four-sample deadline	60.430	2.383	.952	.785
Firm, minimum two samples	60.430	2.383	.952	.785
Firm, recoverable share .8	90.000	0.000	1.000	.500
Silo, recoverable share .8	86.120	1.000	.980	.700
Firm, correct payoff 500	374.065	3.294	.934	.843
Firm, expiration hazard .20	50.000	0.000	1.000	.500

Accuracy is conditional on completion. Payoff and completion include every opportunity. Relative to the silo policy, a one-sample deadline raises total payoff by 3.598 despite lowering completed-case accuracy. An immediate deadline lowers payoff to 50. A mandatory second sample can reduce value by making the intermediate work compulsory while leaving further continuation endogenous.

The executed legacy script independently checks a three-period payoff/time frontier against backward induction, verifies continuation-set nesting on a grid, and checks the nonmyopic counterexample and final-period derivatives. It reports 12,002 checks. These results neither identify firm behavior nor establish a novel single-case index.

The fixed-policy delay-price file separates gross reward less acquisition expense from active-period exposure. Its break-even comparison for the frozen one-sample and silo policies is approximately 3.432 delay-cost units. Those bounds hold the full signal/reward/expiration model and implemented policies fixed. Reoptimizing at each price answers another question.

The horizon is a substantive finite assumption, not an unverified infinite-horizon approximation. The R1 main model instead has exactly two cases, one experiment each, and two slots.

Appendix D. Correlated two-case/two-slot extension: complete details

D.1 Probability space and filtration

The latent state vector takes values (0,0),(0,1),(1,0),(1,1) with public prior mu. Each case has one potential binary signal, conditionally independent given that vector, with likelihood q_i for a correct signal. There are two potential independent survival coins for each case, independent of the states and signals. A survival coin is used only when that case is unresolved across that slot. The potential signal is observed only when its experiment is conducted and its case survives the corresponding slot. Unused draws integrate out. No latent outcome of an irrevocable terminal action is revealed during the window; realization is settled later. This is essential to avoiding unmodeled free information from action outcomes.

The filtration consists of all observed research assignments, stops, survival events, and delivered signals. Expiry itself is uninformative about latent types. For any nonzero-probability observed signal y from case k,

$$\mu_{ab}^{k,y} = \frac{\mu_{ab} \Pr(y \mid \theta_k)}{\sum_{a',b'} \mu_{a'b'} \Pr(y \mid \theta'_k)}.$$

All continuing policies are adapted to this public filtration. At most two experiments occur in total, never more than one on a particular case. Case payoffs remain additive even though their conditional expectations respond to common information.

D.2 Final-slot formula

Fix the current posterior, live set L, and unused-experiment set E. Stopping every case gives $G_L = \sum_{i \in L} g_i$. If k is researched, its own expected resource payoff is $-c_k - d_k + s_k M_{k|k}$, a gain b_k. A passive case i may stop for g_i or remain unresolved across the slot. If retained, its expected payoff is

$$-d_i + s_i[(1 - s_k)g_i + s_k M_{i|k}].$$

Subtracting g_i gives

$$-d_i - h_i g_i + s_i s_k (M_{i|k} - g_i) = a_{i|k}.$$

This accounts separately for expiry of the recipient and nonarrival of the source signal. Their hazards are independent, even though the latent states are correlated. The optimal retention gain is a^+. Summing this and b_k, and taking the best single k or stopping all, proves equation (9). A previously used case may be a recipient and therefore belongs in L even if absent from E. This is the key difference from the independent-case enumeration.

If no experiment is selected in the final slot, waiting creates no signal or action improvement; holding costs and expiry cannot improve nonnegative expected terminal rewards. Thus final idle policies are weakly dominated. All researched and retained survivors choose Bayes-optimal terminal actions after the selected signal, if delivered. No retrospective reuse of a future signal is allowed for a case that acted before it.

D.3 Initial action exhaustion and rational solution

There are four retained subsets R. When R is empty there is one active choice; for each singleton R there are two (no research or that case); when both are retained there are three. The total is

eight. Every case outside R locks its immediate-action payoff and can neither be reopened nor researched later. Every case in R pays d_i for this slot, and a selected experiment pays c_k once whether or not its case survives. Conditional on survivors, if k survived, Bayes-update on its signal; if not, do not update. These are exactly the terms in (10).

The remaining value is explicitly (9), so (10)-(11) are a finite closed evaluation of eight rational expressions with maxima, not an unsolved infinite or general queue recursion. Posterior probabilities and signal probabilities are rational whenever the primitives are rational. The supplied solver uses Fraction arithmetic throughout this calculation.

Under independence, the passive mean $M_{i|k}$ equals g_i. Thus passive research-information retention in the last slot is dominated, including for the previously studied case. The unused case's posterior is unchanged by the first signal. If scheduled second its optimal initial retention gain is the independent w_i, so the eight-action formula reduces to Theorem 1, with initial idle options dominated as already proved. This verifies preservation of the original 137.248 independent example; the correlated model does not replace its assumptions retroactively.

D.4 Complete local game

At one remaining slot, define charged own research gains $b_i - \tau$. If B has an experiment and $b_B > \tau$, A's own best release-and-learn gain is $a^+_{A|B}$; otherwise it is zero. If A also has an experiment it claims research precisely when its charged own gain is positive and weakly exceeds that release gain. A positive tie between early research and later free learning favors its priority claim; a zero tie favors immediate action. If A releases or has no experiment, B researches iff $b_B > \tau$. Given the active case, each passive owner retains iff its own a is positive; otherwise it acts before the signal. These are optimal local actions because passive retention does not alter the active owner's own signal or resource costs. If no case researches, both act.

Let the resulting two components be $L^1_i(\mu, L, E; \tau)$, expressed at full resource costs without subtracting transfers. For a root choice (R,k), the actual component is

$$Q^L_i(R,k) = \mathbf{1}\{i \notin R\} g_i - \mathbf{1}\{i \in R\} d_i - \mathbf{1}\{k = i\} c_i + E[L^1_i(\mu', L', E_k; \tau)].$$

The private component subtracts the root experiment charge and expected future experiment charge, each paid only by the active owner. If a priority toll is applied only when A claims the initial slot, subtract it only from A's private component on that branch. The transfer is never subtracted from total resource payoff.

B's follower alternatives after A claims are its immediate action or retention, giving the set C. If A releases and stays unresolved, B can act, research now, or wait, giving R. If A acts and releases, B has the same three own choices with A removed, giving X. These sets of size 2,3,3 exhaust the eight feasible root actions. Choose B's highest private component in each set, then A's highest private component among the three resulting routes. This is full backward induction of the finite perfect-information local game, including off-path choices. No equilibrium is inferred from a joint-optimal dispatcher policy.

In the full-support synthetic example, at zero charge the three locally optimal route pairs (A resource payoff, B resource payoff) are:

- A claims, B retains: (78.2, 76.689855).
- A releases and stays, B researches: (90.35532, 64.05).
- A acts and releases, B researches: (60, 64.05).

A therefore releases and stays, so the local total is 154.40532. Its claim route has total 154.889855 but is privately worse for A. Even that route does not achieve the efficient 159.69508928 because B declines a later experiment with negative own gain. Adding the R1 marginal-only

priority toll .6405 to the claim route does not change the equilibrium. This computation exposes both initial free-riding and a supported continuation underresearch boundary.

D.5 Signed continuation incentive and full payoff alignment

At a last-slot history with only i's experiment available, the joint gain is b_i+a_j|iˆ+. At a nonnegative fee the active owner will not undertake an experiment with nonpositive own b_i. If the sum is positive despite that, no such fee implements. A conditional credit equal to a_j|iˆ+ makes the owner's gain equal to the joint comparison. A strict credit above -b_i is required for strict research if b_i<0, with equality permitting the prescribed action tie. This proves Theorem 6's exact boundary.

This final-slot credit is not enough to certify the root equilibrium. A's release-and-free-learn payoff and B's retention responses can also change. The full finite-game solution above must be recomputed under any proposed transfer scheme, including future credits. The paper does not claim that the last-slot credit alone attains the optimal root value.

For the stronger theoretical team-payoff contract, let realized resource payoffs, including all own costs and terminal reward, be X_A and X_B. Give i a signed additional internal transfer X_j-K_i, where K_i is an initial fixed constant independent of subsequent choices and signals. Its private total becomes X_A+X_B-K_i. At every public decision, maximizing conditional expected private payoff is maximizing the same total resource value. The finite game's backward induction then selects the joint-optimal feasible continuation at every history, and hence (9)-(11). All eight root choices and all final retention plans are attainable through the defined sequential moves, so no central-feasible action is excluded by the priority right when both owners share this objective.

This is an elementary sufficient alignment result, not a new private-type mechanism. Transfers may be positive or negative, need not balance within owners, and require observable eventual payoffs and authority over internal marginal incentives. Adding both transfers to social payoff would be wrong: the firm objective is still X_A+X_B, not twice that quantity. A fixed constant changes distribution, not decisions. These assumptions are stronger than those of the restricted one-time priority fee, and the construction is not asserted to satisfy the published dynamic-pivot participation or truthfulness conditions under correlated private information.

D.6 Exact example and independent enumeration

For mu=(.58,.02,.02,.38), q_A=.8, q_B=.95, c_A=1, c_B=30, h_A=h_B=.01, S_A=S_B=100, and zero d and r, all state and relevant signal branches have positive probability. Conditional on A surviving, its signal is 1 with probability 11/25 and 0 with probability 14/25. After signal 1 the posterior is (29/110,1/110,2/55,38/55). After signal 0 it is (29/35,1/35,1/140,19/140).

If both survive and the first signal is 1, the final research plan keeps both unresolved and researches B. At this posterior, g_A=800/11, g_B=70. B's own gain is -119/20, while A's cross-information gain is 9401/550. The total increment is 12257/1100. Expected final case payoffs are 4491/50 for A and 1281/20 for B, totaling 15387/100. With a first signal 0, both act for a total 1185/7. These values include the final-slot costs and hazards.

The six first-slot branches have probabilities 1/10000 (neither alive), 99/10000 (only B alive, no delivered A signal), 693/125000 (only A alive, signal 0), 1089/250000 (only A alive, signal 1), 68607/125000 (both alive, signal 0), and 107811/250000 (both alive, signal 1). They sum to one. Weighting the final values and subtracting first-slot acquisition/holding costs gives 249523577/1562500 exactly. The root comparison with every other retained-set/active choice is supplied in JSON and the main table.

The independent enumerator pre-draws physical latent states, both potential signals, and both survival coins per case. It evaluates actual terminal correctness on each path, using a separately calculated likelihood-based action at the information available when that action is locked. The evaluator does not call the rational final-slot formula or root solver. It enumerates all root actions and all observable-history contingent final plans, including retention of the already researched case. Final idle plans are excluded only by the analytic dominance proof; root idle plans are enumerated.

For positive-probability interior branches the number of final schedule/retention maps per root action, in order empty, idle A, active A alone, idle B, active B alone, idle both, active A with both, active B with both, is 1,2,1,2,1,20,18,18, totaling 63. For idle both, both-live states allow five plans (stop; A active with B stopped or retained; B active with A stopped or retained), each singleton survivor allows two, giving $5*2*2=20$. For active A with both retained, each both-live signal node allows three plans and the B-only survivor node allows two, giving $3*3*2=18$. Other nodes have no remaining experiment or no case. Branches of zero probability reduce counts. Bayes-optimal terminal actions are imposed, not counted as distinct deliberately mistaken policies.

There are exactly three named checks, with 63,30,63 maps, totaling 156 output rows. They are the full-support correlated example, a perfect-correlation zero-expiry boundary, and the product prior with the same marginal probabilities and other primitives. Both all-root comparisons and implementation boundary checks are executed, 37 in total. This is a finite exact calculation and enumeration, not a Monte Carlo simulation or a larger queue study.

Appendix E. Two-credit implementation: exact regional proof

E.1 Fixed institution, no omitted payments

This appendix proves Theorems 7-8 on I=[17/100,23/100]. It retains Appendix D's latent binary states, public signal histories, independent .01 expiration, one experiment per case, local priority rights, and immediate-action options. C is earned by conducting initial A research, not by retaining both or by successful survival. B's credit requires that A was initially researched, both survived the first slot, A's delivered signal was 1, and B conducts its final experiment. The payment is earned before its second survival coin. No credits arise after initial B research, initial idling, or terminal action. Credits are sunk once earned and do not change subsequent terminal-action preferences. No experiment can be repeated to collect another credit.

The zero-own-gain research/action tie selects action. A positive initial claim/release tie selects its priority claim, consistently with the original local institution. These are different ties. Conditional Bayes-action ties can use action 1 without affecting expected payoffs. Weak implementation means the outcome chosen by these conventions; strict supported implementation means strict root and discretionary on-path research/retention incentives. It does not require every unused off-path option to have a strict ranking.

E.2 Joint policy and complete exact interval certificate

Write s=99/100 and h=1/100. The target policy has the fixed research/retention decisions in Section 8.4. Terminal actions are also fixed on I: if only A survives its first experiment, act according to that signal; after both survive signal 0, both choose action 0; after both survive signal 1 and B is studied, both surviving cases choose B's signal, except that A uses its initial signal if B expires and no second signal arrives. If A expires initially, B's one experiment uses the prior and its own signal. All these terminal choices are Bayes optimal throughout I. This follows by comparing the unnormalized masses of each binary state after each observed signal. The only two-signal Bayes-action switch inside I is at 41/225 for A after (Y_A,Y_B)=(0,1), a

signal pair the target policy never generates because it stops after Y_A=0.

The target's fixed physical resource components can be derived without posterior division. The relevant joint probabilities are

$$\Pr(\theta_A = 0, Y_A = 0) = .48, \quad \Pr(\theta_A = 1, Y_A = 1) = .32,$$
$$\Pr(Y_A = 1, Y_B = \theta_A) = .202 + .9\delta, \quad \Pr(\theta_B = 0, Y_A = 0) = .336 + .6\delta.$$

Consequently

$$X_A = -1 + 100\{sh(.8) + s^2(.48) + s^3h(.32) + s^4(.202 + .9\delta)\},$$
$$X_B = sh(64.05) + 100s^2(.336 + .6\delta) + s^2(.44)(64.05). \tag{E1}$$

The A cost is paid once even if it expires. B's research payoff 64.05 already includes its final acquisition cost 30 and survival-weighted reward .99(95). The B-only branch uses that value, while the both-surviving signal-1 branch has probability s^2(.44). Thus neither cost nor lost opportunity is counted twice. Expanding (E1) gives (14), so the target value is affine in delta.

For completeness, the maximum physical payoff at every root choice is certified exactly at both interval endpoints:

Retained set, initial assignment	delta=17/100	delta=23/100
Neither, none	120	120
A only, idle	68709/500	68709/500
A only, A	691/5	691/5
B only, idle	246819/2000	246819/2000
B only, B	2481/20	2481/20
Both, idle	36853237179/250000000	38582309997/250000000
Both, A	30486421447/200000000	161147685689/1000000000
Both, B	729696081/5000000	312339/2000

These fractions are finite sums over latent states, potential binary signals and the four potential independent survival coins. At each root the evaluator enumerates every final schedule/retention map, with terminal Bayes actions evaluated separately from the joint solver. There are 63 maps per prior. At both endpoints and the witness, the target is the unique maximizing map; the closest distinct map is lower by 8019/200000=.040095. This is the loss from omitting the worthwhile B-only continuation on its positive-probability survival branch.

To prove the continuum, fix any feasible pure physical policy, including its terminal actions. Its expected payoff is affine in delta because the joint prior is affine and signal/survival likelihoods are fixed. The maximum over terminal actions or any collection of policies is therefore convex. For any competing root choice, its optimized payoff is convex, while the target payoff (E1) is affine. The target strictly exceeds it at both endpoints by the table, hence strictly exceeds it throughout the closed interval by the defining convexity inequality. The same reasoning compares every distinct final schedule/retention map under the target root: its terminal-action-optimized value is convex, and the target exceeds each at both endpoints. This proves strict optimality of the target map on I. No extrapolation from a sampled grid or unproved monotonicity is used. Appendix D's final-idle dominance removes only decisions that cannot improve a payoff; root idle choices are explicitly included.

E.3 All final-slot local continuations

After initial A research and survival of both cases:

1. If Y_A=1, B's own uncredited research gain is -K. Its gain including the observable-history credit is kappa-K. The zero-gain tie selects action, so research occurs exactly when kappa>K. A's passive cross-information gain is (4410450 delta-594259)/22000, strictly positive on I, and it strictly retains if research occurs. Otherwise both act. Joint research is strictly valuable because the sum is (1410450 delta-65159)/22000>0.
2. If Y_A=0, B's own gain is -3(5000 delta-189)/140<0 and no credit is eligible. Thus it acts, and A acts as well. A's hypothetical benefit from a B experiment cannot itself make B conduct it.

If A expires but B survives, expiry is uninformative and B researches at gain 81/20. If only A survives, no experiment remains and it acts using its delivered signal. If neither survives, there is no choice or further payoff. All six positive-probability first-slot histories are thereby covered. Payment C was already earned and is irrelevant to all these continuations.

Off the target route, after initial B research, no credit is ever eligible. If both survive B's signal 0, A's own research gain is -2(225 delta+236)/295<0, and both act. After B's signal 1, A researches exactly when delta<delta_0. On this profitable branch its own gain is (7831-45000 delta)/205, and B acts before the experiment because its passive gain is -38/41. At delta_0 A's zero gain selects action. For delta>delta_0 research remains unprofitable; beyond 41/225 the uncredited own gain has the expression -(450 delta+287)/205, also strictly negative. B's passive cross gain after its signal 0 is -57/59. If B expires and A survives, A researches with uncredited gain 91/5. If only B survives its own first experiment, it acts. These are all B-first survival/signal continuations.

After initial idle retention of both cases, both experiments remain. If both survive, A's own immediate research gain is 91/5, whereas releasing to B gives A the larger cross gain

$$a_{A|B}(\mu) = \frac{99^2}{100^2}100[(.518 + 1.8\delta) - .6] - .6 > 91/5.$$

The inequality holds at 17/100 and increases thereafter. B's own gain is 81/20>0, so A releases, B researches and A retains. The own/passive components are precisely those of the local game, not the joint-optimal final assignment. If only A or B survives, that case conducts its own profitable experiment. With a single case retained initially, the same singleton continuation applies; a case researched initially cannot repeat research. With no live cases, nothing occurs. This exhausts initial-idle, singleton and empty-root continuations as well as both research roots.

For transparency about unused ties: the Bayes-action tie at 41/225 is on an unchosen two-signal route; A's hypothetical retention tie after initial Y_A=0 and a subsequently conducted B experiment is at delta=137947/735075, because the cross gain above 41/225 is 3(735075 delta-137947)/14000. B never chooses that uncredited experiment on I. Neither tie changes (17) or funding. A stronger requirement that every conceivable off-path action and terminal Bayes choice be strict would fail at these points and at delta_0 without further instruments. The stated strict supported implementation deliberately does not claim that stronger property.

E.4 Root follower choices, including failed-continuation credits

If A initially claims, B can act for 60 or retain. Let

$$Y_B^0 = 50.815215 + 117.612\delta.$$

This is its resource/private payoff from retention if it does not conduct research on the both-live signal-1 branch. It includes the uncredited B-only experiment after A expires. It is strictly

above 60 throughout I. If kappa>K, its retained private payoff is $X_B + p\kappa = Y_B^0 + p(\kappa - K) > 60$. If kappa<=K, no eligible experiment occurs and the retained payoff is Y_Bˆ0>60. Thus B strictly retains for **every** pair of nonnegative instruments, not only after imposing the target policy. This also proves the retention slack at the witness's limiting credit.

If A releases but stays unresolved, B may (i) act immediately, leaving only A to wait, (ii) conduct its experiment with both retained, or (iii) wait with A. Its respective own payoffs are

$$60, \qquad 64.05, \qquad .99(64.05) = 63.4095.$$

The third value follows from the actual no-signal final local game just derived, including the cases in which A expires first. It is not inferred from a centrally imposed continuation. B strictly chooses (ii). If A acts and releases, B has exactly the same own three values and again researches initially. These comparisons fully resolve the follower sets C,R,X of Section 8.2, including idle and immediate-action alternatives.

A's private payoffs for claim, release-and-retain, and act-and-release are therefore

$$\begin{cases} (X_A + C, R_A, 60), & \kappa > K, \\ (78.2 + C, R_A, 60), & 0 \le \kappa \le K. \end{cases} \tag{E2}$$

The second branch exists only when K is nonnegative. Without the later B experiment, A's research payoff is .99(80)-1=78.2. In both branches B's root retention is optimal. Since R_A>60 and R_A>78.2 on I, the actual local equilibrium is A-first exactly when C is at least R_A minus its corresponding research component in (E2); otherwise it releases and waits. A's act-and-release option is feasible but strictly dominated by release-and-retain here. This solves root behavior also for instrument pairs that fail implementation.

To derive R_A, if B expires initially and A survives, A researches for 78.2. If both survive, without A's further experiment it acts according to B's signal, whose unconditional correctness probability is .518+1.8 delta. After B's signal 1, its possible own profitable research adds the positive part derived above. Hence

$$R_A = sh(78.2) + 100s^2(.518 + 1.8\delta) + s^2(.41)[(7831 - 45000\delta)/205]^+,$$

which is (18). Subtracting (E1) yields the two pieces (16). The low piece has positive intercept and slope; the high piece is increasing and agrees with the low piece at delta_0. F is positive everywhere on I. Combining (E2) with B's strict research condition proves both directions of (17), not merely a sufficient deviation check. Once the continuation is induced, kappa has no further effect on A's resource component, so even an arbitrarily large continuation credit cannot replace C. A nonnegative initial toll makes its strict free-riding preference stronger.

E.5 Funding constraints and endpoint interpretation

The payment random variable on the implemented policy is $C + \kappa \mathbf{1}_{\mathcal{H}}$, where $\Pr(\mathcal{H}) = p = s^2(.44) > 0$. The event is research after the first signal-1/both-survive history; it does not require either second-slot survival. Both events of receiving C alone and of receiving C+kappa have positive probability. Thus (19) is exact, including the payment after initial A expiry. No eventual realized reward need be observed for settlement.

Put $k = K^+$. The implementation set is the product of C>=F and the half-line kappa>=0,kappa>K; for strict supported implementation replace the first inequality by C>F. Consequently the weak expected/pathwise infimum functions are piecewise F below delta_1 and F+pK/F+K at or above delta_1, with F itself changing slope at delta_0. At delta_0 the two F formulas coincide; kappa=0 is feasible because K<0. At delta_1, kappa=0 fails by the zero-gain

action tie, so the continuous infimum ceases to be attained. At both closed endpoints of I the respective formulas and tie conventions apply without taking an unstated open neighborhood. No claim is made just outside I or at independence, where the target pattern need not persist.

For weak implementation, a single cap equal to its infimum is feasible exactly when delta<delta_1. For strict supported implementation no equality cap is feasible anywhere on I. Under simultaneous expected/pathwise caps, both corresponding slack or weak inequalities are required. Sufficiency follows by choosing sufficiently small increments in each payment, rather than treating an expected limit as permission to exceed a pathwise limit. Since both outlays are increasing in C and kappa, there is no unaccounted trade-off permitting a smaller initial payment in exchange for a larger continuation payment after the efficient branch is induced. These are gross funding requirements for this specified class, not firm resource costs or minimum funding over all possible mechanisms.

P.1 Status, unit, and estimands before methods

No partner, data agreement, approved treatment, recruitment, observations, or experiments exist. This is a prospective protocol. The theoretical model has two one-experiment cases and two slots, with independent and correlated binary-state specifications. The study tests its shared-capacity mechanism in eligible operational queues, not its literal numerical calibration.

Primary unit: an analyst-queue episode with a frozen intake cohort and a defined capacity window. A queue consists of all cases sharing the same analyst/reviewer resource. An episode can contain more than two cases; theoretical optimality is not asserted for that larger queue. A matched two-case substudy is possible only if the institution can document a bounded two-slot decision window and reserve capacity without displacing uncounted cases. Cases excluded from the cohort but affected by reserved capacity must be followed as spillovers.

Primary estimand: intention-to-treat effect of offering an opportunity-loss scheduling aid on total 90-day net payoff summed over the episode's registered cases, averaged equally across a crossed evidence-provision assignment. The primary target is the mean episode total at the prespecified cohort definition, not a mean among completed purchases. Report case-weighted results secondarily. Unequal episode size and any weighting are frozen before assignment.

If defensible monetary bounds are incomplete, the primary estimand is interval valued. Declare a favorable welfare effect only when its lower confidence bound is positive across the prespecified admissible valuation set. A faster queue with worse total expected payoff is not a successful treatment.

Secondary confirmatory family, with Holm adjustment across seven contrasts: evidence provision effect on total payoff; aid-by-evidence interaction on total payoff; aid effect on all-worker analysis minutes; aid effect on summed unresolved case-hours through day 90; aid effect on cases completed by day 30; aid effect on expired cases; aid effect on total rework and switching expenditure through day 90. Case reversal rates and mechanism interactions are exploratory and explicitly labeled.

P.2 Feasibility and eligibility

Seek a partner with routine, non-safety-critical procurement decisions, reliable intake for purchases and rejections, logged evidence requests, a shared analyst roster, verified quote expiry, and linked operational outcomes. Exclude safety-critical, legally mandated, personnel, emergency, and strategic high-value reviews. The partner defines exposure caps, authority limits, and reversible-action terms before enrollment. No such caps are invented here.

A read-only feasibility audit must establish: the actual shared-resource boundary; how

managers currently claim priority; whether immediate action really avoids analyst use; packet production durations; existing mandatory reviews; rejected/unresolved intake coverage; capacity window commitments; and spillovers to adjacent queues. If terminal approval itself uses the same scarce reviewer, the two-slot research-only model does not fit without redesign. Record that mismatch rather than silently treating both services as the same.

Public BPI Challenge 2019 organizer documentation verifies a procurement-process event-log candidate. Repository/file access and license were not verified; no schema or rows were analyzed. It lacks a verified research-queue mapping in this work. It may inform process measurement if later accessed under its actual terms, but cannot identify the treatment effect without intake, resource links, and economic outcomes.

P.3 Design and treatments

Use a parallel 2-by-2 design at the whole independent analyst-pool level. If too few pools exist, consider randomly assigned pool-period blocks only after demonstrating feasible washout and tracking all carryover cases; this changes assumptions and requires a new registration. Do not randomize individual cases independently in one shared queue and then assume no interference.

Evidence factor: provide one frozen incremental packet for each eligible case, versus existing information. Packet production time, source costs, and worker effort count in the resource ledger. Keep purchasing alternatives, legal information, price opportunities, and formal authority comparable. Display design is held constant conditional on provided facts. A compressed-display substudy, if later chosen, must preserve accessible content and has an attention estimand rather than an acquisition estimand.

Scheduling factor: offer a queue-level opportunity-loss aid versus existing priority practice. At a review point, the aid records for each case: current best action; available incremental experiment; credible expected action improvement; acquisition time; option to act immediately; consequences of waiting; quote expiry; recoverability; and the identity of other cases displaced by the next analyst slot. It distinguishes a research value estimate from a deferral-loss estimate. It displays a proposed research order and explicit act-now or retain-for-later decisions, with documented overrides.

Record prespecified cross-case prior links, signal-source identities, posterior revisions in both cases, and whether an already researched case remains unresolved to receive another case's signal. Do not infer independence merely from separate case records. Correlated episodes require a joint prior and signal model; applying the independent u-score to updated marginals is not justified. The final-slot information credit is a theoretical incentive calculation, not an authorized transfer treatment.

In verified independent two-case one-packet episodes, the aid can calculate b, w, and u under transparent parameter inputs. In larger queues it is only a documented scheduling recommendation, not a theorem-backed globally optimal index. Uncertain inputs must produce sensitivity ranges, not a falsely precise rank. Record when ranks are indeterminate over the full uncertainty set.

The offered aid does not implement real marginal incentives automatically. No effect is labeled a toll-mechanism treatment effect. If management later authorizes actual internal charges, register a distinct arm specifying who pays, when, whether later service is uncharged, whether the charge changes the marginal budget, and whether the analyst's priority right is enforceable. A generic charge per minute is not the posted first-slot charge in Theorem 3. No actual charge, binding deadline, or authorization change is part of the present study.

P.4 Interference and identification

Treatment directly changes delay on other cases; that spillover is the mechanism. The primary total episode payoff includes it. Randomization identifies the policy-offer effect if cohort registration is unaffected by assignment, outcomes are measured comparably, and cross-pool spillovers or carryover are either negligible or included in a prespecified exposure design.

Use staffing and resource rosters to identify shared analysts, reviewers, vendors, stock constraints, and work redirected outside the pool. Log both treated and displaced cases, including unserved requests. Estimate exploratory effects by exposure to treated competing cases, but do not call those causal direct effects without a registered interference model. If resource sharing is organization-wide, the independent-pool design may be infeasible. The correct response is redesign, not inflated apparent sample size.

Analyze with a saturated factorial regression or assignment-based contrast at episode level, retaining randomization strata and frozen baseline load, case mix, and holding exposure covariates. If episodes repeat within pools, inference clusters at the assignment pool. Use randomization inference consistent with the actual assignment, with small-cluster robust inference as a companion. Define the aid main contrast as equally averaged across the two evidence levels.

Do not control in the main regression for post-treatment completed-case status, evidence requested/opened, research minutes, priority rank, or elapsed time. These are mediators/outcomes. Pre-treatment difficulty measures include novelty, baseline available evidence, stakes, number of alternatives, resource requirement, approval layers, and time sensitivity.

The mechanism prediction is that changing priority rights or accounting for a rival's deferral loss matters most when local and joint orderings disagree. Define disagreement using frozen pre-treatment inputs and a documented valuation uncertainty set. Larger effects in queues with split budgets or severe load are descriptive heterogeneity unless those features are randomly manipulated. A future randomized display of verified competing-case costs could sharpen a mechanism test while holding the focal packet fixed.

P.5 Data dictionary and timing

Variable	Definition and units	Source/timing	Main concern
pool_id	All cases sharing analyst capacity	Baseline roster	Hidden shared workers
episode_id	Frozen cohort and capacity window	Intake before assignment	Selective episode construction
case_id	Unique decision, bundled lines resolved	Intake	Split/duplicate cases
eligible	Pre-treatment inclusion rule	Intake attributes	Strategic exclusions
assigned_E, assigned_S	Evidence/aid assignments	Randomization ledger	Offer versus uptake
competing_ids	Cases displaced by a requested slot	Resource ledger	Unrecorded opportunity loss
priority_owner	Existing first-slot right and override authority	Baseline process	Informal authority
analyst_available	Research capacity by slot and worker	Staffing log	Capacity altered by treatment

Variable	Definition and units	Source/timing	Main concern
request_time	First research/decision intake	System log	Backdating
research_start_end	Work on each experiment	Work log/time sample	Offline work
passive_exposure	Unresolved waiting time without own research	Derived log	Calendar time versus active effort
action_time_type	Purchase, reject, expire, or unresolved	Authorization/terms	Missing rejections
expiry_time_terms	Verified quote/opportunity expiry and extension	Vendor terms	Endogenous extension
prior_risk	Frozen uncertainty assessment	Pre-treatment rubric	Unvalidated probabilities
packet_quality	Likelihood/precision evidence where estimable	External historical validation	Availability not use
stakes_recovery	Documented action loss and recoverable share/cost	Baseline contracts	Recovery treated as free
all_worker_minutes	Focal and displaced workers' active time	Linked work sampling	Unobserved displacement
terminal_net_value	Realized action margin before other exclusive costs	Finance/operations	Missing unchosen outcome
rework_switching	Downstream corrections, reversal and replacement cost	90-day follow-up	Delayed posting
residual_holding	Cost schedule not counted in margins/expiration	Independent costing	Double counting
cost_bounds	Justified lower/upper costs and dependencies	Before outcomes	Favorable valuation selection
unresolved_90	Registered case still open at day 90	Follow-up	Completer selection
episode_bounds	Sum of all case and resource payoff bounds	Derived ledger	Omitted cases and external work

This is a proposed schema, not verified columns from a firm dataset. Independent validation is needed before treating stated confidence as a calibrated prior or a packet's quality as a likelihood. Data availability, precision, actual evidence use, capacity rights, and outcomes are distinct variables.

P.6 Welfare ledger, missing outcomes, and bounded costs

Each resource cost and consequence has one ledger category. Acquisition expense counts actual production and analyst work once. Holding exposure counts residual lost capacity or operating costs not already counted in action margin or expiration losses. Expired cases have no realized opportunity payoff under the literal model, but real organizations may retain salvage; the field ledger must use verified institutional terms instead. Transfers for hypothetical internal priority charges are excluded from organization-wide payoff.

The theoretical externality u_j is not added to observed downstream loss. It is an opportunity-cost comparison used to schedule; realized policy payoff already includes the rival case. Adding both would double count. Likewise, a nominal time saving is not necessarily cash savings; report physical minutes and scarce-capacity shadow-price bounds separately.

At a fixed implemented policy, mean payoff is evaluated for all enrolled cases over the same horizon. Random assignment identifies total realized policy value without observing every unchosen supplier's quality. It does not identify decision-specific regret relative to an oracle. Common independent ex post market information may support a secondary quality measure only under a validated mapping to economic outcomes.

Report unresolved and missing-payoff fractions by arm. Never compare mean delay only among completed cases. Sum restricted case-hours through the fixed horizon, with explicit rejection treated as resolution and expiration as a competing event. If missing case payoff is institutionally bounded by [L_i,U_i], use those bounds. Without defensible limits, do not assign zero or simulate informative losses. Report worst-case mean contrasts and sensitivity to missingness. MAR imputation may be secondary only with assumptions stated.

For fixed policy z and known non-delay primitives, uncertain residual costs give $W_z = A_z - \sum_i d_i T_{iz}$. Endpoint bounds hold that policy and the full signal/reward/expiration model fixed. They do not establish robustness to model error. With uncertain priors, recovery, precision, or hazards, define a full joint uncertainty set with documented dependencies and recompute every candidate policy contrast. Require uniform nonnegative lower comparison for a certified ordering; otherwise report ambiguity. No such real-data uncertainty set is currently estimated.

P.7 Precision and feasibility planning

Illustrative arithmetic only: with 120 genuinely independent analyst pools, 20 episodes per pool, balanced four-cell assignment, and episode-payoff ICC .05, the design effect is 1+19(.05)=1.95. The normal-approximation 80% power, two-sided 5% aid-main-effect MDE is $(1.96 + .84)\sqrt{4(1.95)/2400} = .160$ episode-payoff standard deviations. The corresponding factorial interaction MDE is .319 SD. At ICC 0 and .10, main-effect MDEs are .114 and .195 SD. These are hypothetical pools and episodes, not observed sample sizes, variances, or power estimates.

The independent assignment unit must be real. Twenty cases in one episode are not twenty independent queue experiments, and repeated episodes with the same analyst require clustered inference. Unequal episodes, fewer pools, carryover, uptake failures, and wide payoff bounds worsen precision. Final power requires a partner's baseline outcome variance, ICC, assignment structure, and economically meaningful currency or capacity effect, followed by simulation under the actual design. None is available.

A feasibility pilot measures only nuisance quantities and operational limits, not a treatment-effect-driven outcome choice. Freeze final enrollment, cohort rules, hypotheses, valuation bounds, and analysis dates before examining treatment outcomes. No unadjusted repeated p-value peeking is planned. Any confirmatory interim analysis requires a separately registered alpha-spending or always-valid method.

Go/no-go gates: complete intake; real independent capacity pools; auditable priority rights; comparable alternatives and capacity; bounded stakes; linked all-case outcomes; displaced-work coverage; and institutional authorization. Failure triggers redesign rather than a nominal larger N.

P.8 Unpopulated reporting template

Assignment	Registered episodes	All cases	Unresolved fraction	Total minutes	Payoff lower/upper
Standard packet, local order	Not observed	Not observed	Not observed	Not observed	Not observed
Added packet, local order	Not observed	Not observed	Not observed	Not observed	Not observed
Standard packet, queue aid	Not observed	Not observed	Not observed	Not observed	Not observed
Added packet, queue aid	Not observed	Not observed	Not observed	Not observed	Not observed

No empirical estimates, assignments, p-values, or implemented organizational outcomes are supplied. Synthetic model results belong to the theory section only.

The two-credit result is theoretical only. A future feasibility audit could distinguish expected internal credit authorization from gross pathwise funding capacity and verify that research and the specified public signal history are observable. It supplies no authorization to pay credits or add an experimental treatment, and no real transfer data exist.